

Cast Gold Restorations

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The cast gold restoration has lessened in popularity over the past 20 years because of the increased emphasis on esthetics, but it remains an excellent restoration with a long history of success. If used with care, gold alloy is considered to have the greatest longevity of any restorative material used in dentistry. This opinion is generally supported by longitudinal studies,[1–3] although it is disputed by other studies.[4–6] Cast gold may be used for intracoronal (inlays) or extra-coronal (complete-coverage crowns) restorations, as well as for restorations that are a combination of both (onlays or partial-coverage crowns) (Fig 20-1).



Fig 20-1 *(a)* Inlay, partial-coverage, and complete-coverage cast restorations. *(b)* Partial-coverage cast restoration (mesio-occlusodistal [MOD] onlay).

Several long-term studies have demonstrated the excellent clinical behaviour of cast gold restorations over time.[7–9] Restoration longevity runs from an average of 10 to 20 years and up to 34 and 52 years in selected practices.[10–12] The survival rate ranges from 87% to 97% at 10 years,[7,9,13] 87% to 90% at 20 years,[9,13] and 73.5% to 98% at 30 years.[8,9,13] Compared with all-ceramic inlays and onlays, cast gold restorations have proven more effective under mechanical stresses such as bruxism.[14]

The failure pattern of these restorations can be divided into biologic and technical reasons. The most prevalent biologic reasons are, in order: secondary caries, tooth fracture, endodontic treatment, primary caries, and periodontal involvement. Loss of retention is the main reason for technical failure, followed by extensive wear.[13,15,16]

Gold castings present several advantages over direct restorative materials such as silver amalgam or resin composite. Because castings are fabricated using an indirect technique, it is possible to achieve nearly ideal contours and occlusion.[17] Gold alloy is a strong material that rarely fractures and, when used as an extracoronal restoration, can provide protection to the tooth.[18–20] Other materials such as titanium and nickel-chrome alloys have shown clinically acceptable results, but gold alloy remains the dental casting material of choice.[21] Gold wears at a rate similar to that of enamel, so it does not cause accelerated wear of the opposing teeth.[22] It casts easily and accurately, and if a type I or II gold is used, marginal adaptation can usually be improved after the restoration is cast. Gold is also resistant to corrosion.[23]

Allergic reactions to gold are relatively rare but have been documented. In a study of 200 patients with persistent oral or cutaneous lesions and suspected gold allergy, 8.5% were found to have a reaction to patch testing with gold salts. Still, gold is considered among the most biocompatible of dental materials.[24]

Gold shows better resistance to at-home vital bleaching agents, such as 10% carbamide peroxide, compared with unpolished amalgam and nickel chrome.[25] However, galvanic corrosion has been shown to cause unpleasant, sometimes painful, transient effects when gold alloy is in direct contact with zinc-containing amalgam restorations. As passivation processes develop, naturally or by actively brushing the new amalgam restoration with tin oxide, side effects subside.[26–28]

The primary drawback to the use of cast gold rather than direct restorations is higher initial cost, because castings require at least two appointments for the patient and have associated laboratory costs. However, the long-term success of cast gold

restorations offsets this initial investment and decreases frequency of "remakes," preserving tooth structure, pulp vitality, and subsequent expense. For esthetic reasons, gold castings are usually used to restore posterior teeth that are not in the esthetic zone. Preparations for anterior teeth are usually designed so that the gold is not seen during speech and smiling. Esthetics may also be preserved in maxillary posterior teeth when cast gold restorations are used (Figs 20-2 and 20-3). Postoperative tooth sensitivity following insertion of cast gold restorations can usually be prevented with the proper use of sealers, liners, and bases.[29,30] This chapter addresses the indications, materials, and clinical steps for the fabrication of cast gold restorations.



Fig 20-2*(a and b)* Cast gold inlays that preserve tooth structure and are generally not seen during speech and smiling.



Fig 20-3*(a)* Partial-coverage gold restoration. *(b)* When viewed from the anterior, the gold display is minimal, thus satisfying both functional and esthetic demands.

Indications

Traditionally, the indications for a cast gold restoration range from a tooth with a relatively small caries lesion restored with an inlay to a severely weakened and/or malfunctioning tooth restored with a complete-coverage crown. However, direct resin composite or amalgam restorations are more conservative and usually indicated over cast gold for restoration of small caries lesions. While small inlays have been shown to exhibit the highest failure rate of all cast gold restoration types,[9] their long-term clinical success remains excellent. Cast gold restorations may be used to restore teeth with primary caries lesions or to replace existing restorations.

They are generally indicated for situations in which other, less expensive materials are not suitable for establishing proper proximal and/or occlusal contacts, creating appropriate axial contours, or protecting the remaining tooth structure. A cast gold restoration may also be indicated when gold has been used to restore adjacent and/or opposing teeth to avoid corrosion problems arising from the use of dissimilar metals in close proximity in the same mouth.[31,32] A cast gold restoration may be specifically recommended for posterior areas and for alleviating the deleterious effects of bruxism. Reduction requirements for a full-coverage cast gold crown are more conservative than for any other type of full-coverage restoration, and the wear properties of gold make it an ideal choice when there is opposing intact tooth structure.[22]

The morphology of posterior teeth, the number of carious surfaces, the number of restored surfaces, the width and depth of existing restorations,[33] and the occlusal relationships must be considered when determining the need for a cast restoration. Nonworking-side occlusal contacts can be especially important. Hiatt[34] has shown a significant increase in the number of vertical fractures in teeth with nonworking-side contacts.

Basic Principles of Cast Restorations

While there are many acceptable techniques and designs for cast restorations, certain principles always apply.

Conservation of tooth structure

Preparations should be made as conservatively as possible to maximize the remaining tooth structure, decrease the likelihood of tooth fracture, and lessen the possibility of postoperative sensitivity and pulpal pathosis. Tooth reduction should not exceed the amount that will allow adequate material thickness necessary to resist fracture, deformation, or wear during function. From an anatomical standpoint, nonfunctional cusps (buccal cusps of maxillary teeth, lingual cusps of mandibular teeth) of molars and functional cusps of maxillary premolars possess steep inclines. The enamel is also thinner on nonfunctional cusps. The clinician should keep these elements in mind during treatment planning and cavity preparation.[35–38]

Tooth preservation involves more than simply minimizing the removal of tooth structure. Preparations must be designed to protect remaining tooth structure. This may involve additional reduction to remove weak tooth structure, or it may

necessitate cuspal coverage. Studies have shown that as a cavity preparation gets wider[39,40] and deeper,[41] progressive weakening of the tooth occurs. When an inlay preparation exceeds one-third of the intercuspal (buccal cusp tip to lingual cusp tip) width, an onlay or another extracoronal restoration should be considered to protect the cusps of the tooth. Indeed, a conservative three-surface (mesio-occlusodistal [MOD]) restoration has proven to decrease tooth stiffness by 50% and a large three-surface restoration by as much as 78%.[42] Gold onlays or full-gold cast crowns provide a splinting effect and greater resistance to cuspal movement. They virtually double the resistance to fracture when compared with an intact tooth.[42–44] This reinforcing effect is particularly significant in the maxillary premolars.

Two retrospective studies[19,45] concluded that a significant factor in the prevention of catastrophic failure of endodontical-ly treated teeth was the presence of a full- or partial-coverage restoration. Another 9-year retrospective cohort study[46] demonstrated that endodontically treated teeth with intracoronal restorations were six times more likely to fail catastrophically than teeth restored with full-coverage restorations.[19,45,46] Because extracoronal restorations cover both facial and lingual surfaces, they protect remaining tooth structure and reduce the incidence of tooth fracture.[47]

Retention and resistance form

Retention and resistance form are two separate but related features of a preparation. Correctly incorporated, they resist unseating vertical, lateral, and oblique forces that are placed on the restoration during function or parafunction.[48] *Retention form* resists forces attempting to remove a restoration parallel to the path of insertion. *Resistance form* resists forces attempting to dislodge a restoration obliquely to the path of insertion. Retention form and resistance form are usually closely interrelated and may be difficult to distinguish clinically as separate features. However, they both are affected by four factors. The first is the resulting stress quantified by the ratio of applied force to surface area. The force itself varies in its magnitude, frequency, duration, and direction. The other three factors are the final tooth preparation geometry, the luting material type and thickness, and the tooth and crown surface texturing.[47,49,50]

Retention is gained when two or more walls oppose each other. These walls may be intracoronal, extracoronal, or a combination of the two (Fig 20-4). The amount of retention created by these opposing walls is determined by several factors. The degree of convergence toward the occlusal (extracoronal walls) or divergence

toward the occlusal (intracoronal walls) and the length of the walls are the most important factors. One long wall opposed by a short wall provides retention equivalent only to that imparted by the short wall. The longer[51] and more nearly parallel the walls, the greater the retention. The further apart the opposing walls and the greater the surface areas, the more retentive the restoration will be.[52] Certain implant internal connections use these principles, creating an effective self-locking Morse taper configuration. In practice, the taper achieved bears little resemblance to that advocated in textbooks (a combined 6 to 12 degrees), and most teeth are prepared with a taper in excess of 12 degrees with clinical success.[53–55] Although a rapid loss of retention occurs as the taper progresses from 5 to 10 degrees for each wall,[56] a total convergence up to 16 degrees still provides adequate retention.[53,54]

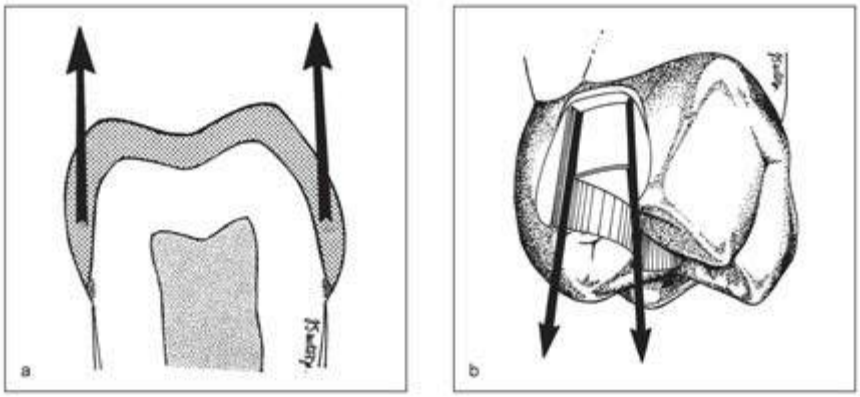


Fig 20-4*(a)* Complete-coverage, or full-veneer, crowns rely on the opposing external walls to provide retention for the restoration. A slight taper of the walls in the occlusal direction provides a path of insertion. *(b)* Inlays (pictured) and onlays rely on tapering opposing internal walls for retention.

At a constant taper, retention increases with increasing preparation length, surface area, and preparation diameter and also with the use of a resin-based cement combined with surface treatments.[47,49,50] It is important to note that the retention of a casting prior to cementation is not an indication of its retention postcementation.[57] The addition of intracoronal features has different effects on retention and resistance. The addition of proximal boxes or pins improves both retention and resistance, whereas proximal grooves and slots/potholes mostly enhance resistance form[58] (Figs 20-5 and 20-6). Posterior teeth that are short may present a problem despite their larger diameter. Short, wide preparations do not resist rotational forces well. Adding a groove or a pin will reduce this rotational radius and provide additional resistance.

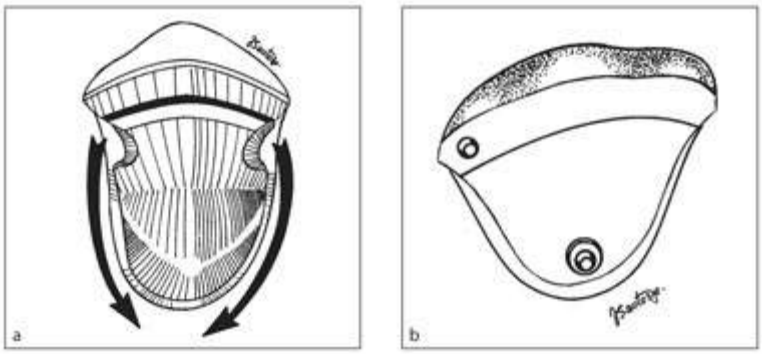


Fig 20-5*(a)* Resistance to lingual and rotational forces may be provided by proximal grooves. *(b)* Pinholes also provide resistance to lingual displacement.



Fig 20-6*(a)* This mandibular molar preparation for a partial-coverage cast gold restoration shows the use of axial grooves and an occlusal dovetail to provide retention and resistance form for the partial-coverage casting. *(b and c)* Note the die and casting showing the same features. *(d)* Definitive restoration.

A recent study[59] demonstrated that the positive influence of grooves or boxes is limited by the height, width, and taper of the existing preparation. The study showed that grooves and boxes seem to possess a limited effect on short, wide, and excessively tapered preparations. The most effective method to improve retention was to decrease the convergence of the preparation's 2.5-mm total height in the cervical 1.5 mm from 20 to 8 degrees. Insufficient retention is a major cause of failure of fixed prostheses, exceeded only by failure due to dental caries and porcelain fracture.[6]

Recent developments in luting agents and techniques of metal conditioning allow bonding of cast restorations to the tooth.[60–62] This greatly improves the resistance to displacement from the preparation. Although less often discussed, the cement's

compression and shear strength as a function of film thickness, along with the tooth-cement and cement-crown adhesive properties, appear to be essential parameters affecting retention and resistance. However, the long-term effectiveness of bonded castings is not yet known.

Resistance form is essential to providing long-term stability of cemented restorations. Taper has a limited influence. A variation of ± 10% taper increases or decreases resistance by ± 10%.[43,44] Indeed, the rationale for resistance form has evolved from a theoretical model[63,64] to a more clinically relevant idea.[44] The theoretical *self-limiting taper concept*, or "on-off" theory, was defined as a definite convergence angle above which all resistance to lateral force application is lost because of the absence of a cement compression zone on the abutment. However, this has been challenged by the importance of the cement's properties, thickness, and mechanical behavior under compression.[50,64,65] Cement thickness should be kept at the minimum level compatible with seating of the restoration. Texturing the surface of the abutment and crown before cementation increases the resistance to dynamic lateral loading.[50]

Pulpal considerations

Tooth preparation for any cast restoration involves dentin and therefore affects the pulp. Often in situations in which a restoration is indicated, the pulp has already been traumatized by caries and/or previous restorations. Prior to tooth preparation, the pulp should be tested for vitality, and radiographs should be taken. Tooth symptoms and any restorative history should be noted. The additional trauma created by tooth preparation can be enough to cause necrosis of an unhealthy pulp. If the vitality of the pulp is questionable, a reevaluation of the pulpal vitality should be performed 4 to 6 months after the initial evaluation. Endodontic therapy is easier and more predictable and economical for the patient if it is performed before the definitive restoration is placed. Endodontic therapy through an existing restoration is difficult and may lead to decreased crown retention, marginal leakage, and recurrent caries.[66] One study[67] determined that the cast gold restoration presented the lowest rate of bacterial leakage when endodontic access openings were closed with either resin composite or amalgam, compared with other types of full-coverage restorations. Even so, endodontic therapy should be performed prior to preparation.

Pulpal considerations for cast restorations are the same as those for direct restorations. The best pulpal protection is a thick layer of sound dentin. A thorough discussion of the subject can be found in chapter 6. The thinner the dentin, the more

permeable and susceptible it is to penetration of bacteria and uncured monomers.[68,69] A pulpal reaction will be triggered if the remaining dentinal thickness (RDT) is less than 1 mm (see chapter 6). In situations in which the RDT is very thin, application of a sealer or liner is indicated, followed by an observation period to determine that the tooth remains symptom-free prior to permanent seating the definitive restoration.[70]

Certain anatomical considerations are critical to planning the preparation, especially in younger patients. The pulp horns may be large enough to be exposed during preparation. Examination of preoperative radiographs is imperative to identify potential problems. Because the pulp is narrower in the cervical region of the tooth, pulpal exposures are less likely to occur in this area.

Postcementation thermal sensitivity is sometimes a problem. Several methods to prevent or minimize this problem have been recommended. Some clinicians scrub the preparations with antimicrobial solutions to reduce bacterial contamination, a possible cause of postoperative sensitivity. Other factors affecting pulpal response include RDT and postoperative time.[70–72] Resin varnish and Gluma desensitizer (Heraeus Kulzer) should be avoided in favor of adhesive primers containing an inhibitor of matrix metalloproteinase (MMP), which seems to enhance the bond durability and act as an efficient antibacterial agent.[73–75]

Gingival considerations

When a tooth is prepared for a cast restoration, finish lines (and therefore casting margins) can be placed supragingivally or intracrevicularly.[76] It is well established that supragingival crown margins are compatible with gingival health, but both configurations have their proponents.[77,78] However, there is general agreement that it is only at the completion of the initial periodontal therapy that the decision on the location of finish lines should be made.[79] Should crown lengthening be necessary, the waiting period before final tooth preparation should provide adequate time for healing and tissue stabilization. Each margin location has its indications, as described below.

Supragingival margins present the following advantages:

- Preparation, impression, and temporization are facilitated.
- The restoration can be finished/cemented and cleaned with ease.
- Patient hygiene measures are more effective, and evaluation at recall is facilitated.

• Incidence of inflammation in adjacent tissues is reduced.

Intracrevicular margins may be warranted in the following situations:

• Existing restorations, caries lesions, or erosion require that the finish line be placed more apically.
• Refinement or modification of an existing finish line is needed.
• Additional retention or resistance form is needed.
• There is a long contact area on a proximal surface.
• The emergence profile requires modification.
• Root sensitivity is unresolved with conservative measures.

Note that esthetics is obviously not a criterion for margin placement in a cast gold restoration.

From a restorative standpoint, subgingival margin placement increases the level of difficulty and reduces predictability. Periodontally, even a well-fitting margin may trigger a pathologic response.[80–83]

The term *subgingival* implies the placement of a restoration margin somewhere between the free gingival margin and the alveolar crest. *Intracrevicular* refers to the placement of a margin within the gingival sulcus and above the junctional epithelium or epithelial attachment (Fig 20-7). Ultimately, the fit, finish, and emergence profile at intracrevicular finish lines may be as significant to gingival health as the location of the finish line in relation to the free gingiva.[84,85]

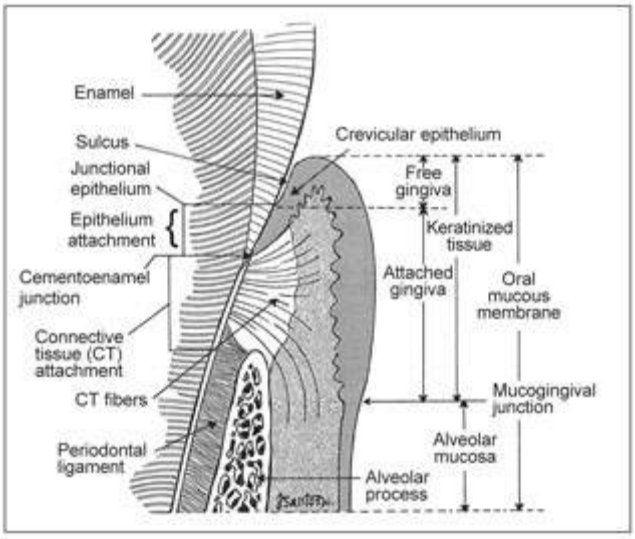


Fig 20-7 Anatomy of the periodontium, illustrating the three zones of the den-togingival junction.[76]

One other controversial issue pertains to restoration margins impinging on the biologic width. Gargiulo's biologic concept and dimensions have been challenged.[86–88] Of the three components of the biologic width, only the connective tissue attachment seems to be relatively constant. Sulcus depth and epithelial attachment show wide variations. Histologically, the sulcus depth is at 0.5 mm, whereas clinically, it ranges from 1.0 to 4.0 mm and is affected by probing force, tooth position, and tissue health status. The dentogingival complex follows human biologic variability. The most predictable method to assess the patient's specific biologic width is to probe the osseous crest.[87]

Biologic width violation can result in an inflammatory response leading to attachment loss, pocket formation, and osseous resorption. Most studies have been performed on animals, and there is actually very limited data on biologic width.[88]

The consequences of restoration margins impinging on the biologic width remain controversial. A 2-year prospective study failed to correlate the violation of biologic width with bone resorption or gingival recession. On the other hand, a statistically significant increase in probing depth and gingival inflammation was observed, but its clinical significance remains uncertain.[88]

It is important to remember that the bone scallop approximately parallels the cementoenamel junction (CEJ) circumferentially.[86–88] Frequently, a preparation finish line on an anterior tooth does not parallel this osseous scallop, thus inducing an interproximal violation of biologic width.

The interproximal bone in posterior areas has a flat architecture and adequate thickness, so a scallop pattern in the restoration finish line is not as necessary. However, a thin labial or buccal plate over a root prominence is not uncommon, so these areas are at risk.[89] A phenotype characterized by a slender tooth and thin gingival tissue may be prone to gingival recession following a biologically or mechanically aggressive procedure.[90] At least 3.0 mm of height of attached gingiva and free gingiva should be present prior to any restorative procedure in which the finish line approaches gingival tissue.[76]

If possible, finish lines should not be prepared deeper than 0.5 to 1.0 mm into the sulcus[91,92] (no closer than 2.5 mm from the osseous crest) or closer than 1.0 mm to the base of the sul-cus. 76 A more complete discussion of this subject may be found in chapter 1.

Because gingival health may be adversely affected by intracrevicular finish lines,[79,82,93–96] finish lines should be placed supragingivally if the situation permits. When finish lines must be placed within the gingival crevice because of caries

lesions, existing restorations, fractures, root sensitivity, or short clinical crowns,[97,98] care must be exercised to minimize the damage to gingival tissues. The fragile soft tissue may be reflected by careful placement of a deflection cord (gingival retraction cord), prior to final preparation of the finish line, to avoid soft tissue damage.

Finish lines

The term *finish line* refers to the border of the preparation where the prepared tooth structure meets the unprepared surface of the tooth. The type of finish line depends on the clinical situation. A smooth, well-defined finish line is beneficial, regardless of the design used, to facilitate laboratory procedures and finishing of the restoration. Selection of the type of finish line may be dictated by the shape of the tooth (bell-shaped versus flat) (Fig 20-8), the desired location of the finish line, or operator preference. The most common types of finish lines for cast restorations are knife-edged, chamfer, and shoulder. Both the chamfer and the shoulder configurations may be beveled or unbeveled (Fig 20-9a).

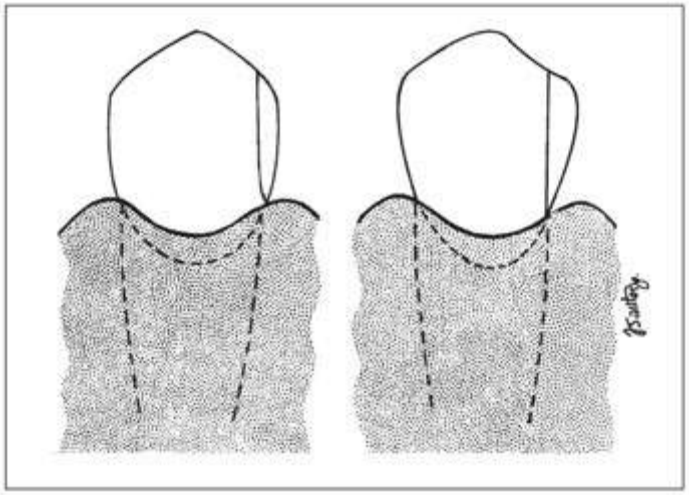


Fig 20-8 Flat proximal surfaces *(left)* allow the easy formation of a chamfered finish line, while bell-shaped anatomy *(right)* makes it difficult to create such a finish line without severe reduction of the proximal surface.

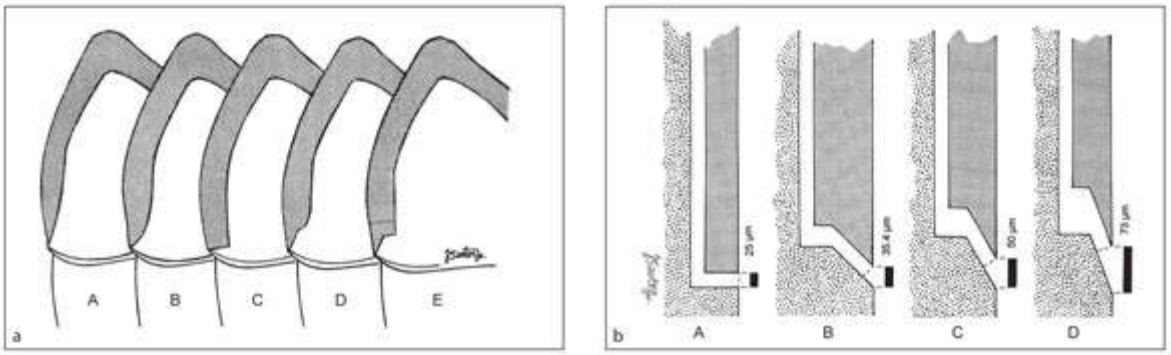


Fig 20-9*(a)* Forms of finish lines: knife-edged (A); chamfer (B); shoulder (C); beveled chamfer (D); beveled shoulder (E). *(b)* Effect of the same cement film thickness on cervical marginal openings, comparing a shoulder marginal configuration with no bevel and beveled shoulders at varied bevel angles: butt margin with a 90-degree cavosurface angle (A); shoulder margin with a 45-degree bevel (B); shoulder margin with a 60-degree bevel (C); shoulder margin with a 70-degree bevel (D). The broken line represents the cement film thickness of 25 μm at the margin. As steepness of the bevel increases, the marginal opening increases.[107]

According to a dental survey, there is no agreement on the ideal finish line.[99] The type of finish line and presence or absence of a bevel will depend on tooth morphology, its location, and the type of restorative material used.

The slip joint versus the butt joint theory

Despite a lack of consensus regarding the maximal allowable marginal gap, a gap between 39 and 120 μm may be considered acceptable.[100–103] It has long been held that the addition of bevels to cast gold preparations helps to reduce marginal gaps.[104] However, this was based on mathematical models that did not take cement thickness into account. As explicitly demonstrated by Ostlund$_{105}$ and supported by others,[106,107] beveling beyond 45 degrees significantly increases the cement-filled opening (Fig 20-9b). Simply put, the steeper the bevel angle, the larger the vertical discrepancy (opening) between the edge of the casting and the finish line. This assumes that the casting is fully relieved, allowing minimum cement thickness. Another study$_{108}$ later demonstrated that crowns with a shoulder finish line configuration had better marginal fit than did those with a beveled finish line configuration. The study did not describe or quantify the amount of difference of internal fit of the castings. Factors such as existing taper, die relief–spacer thickness, laboratory technique, cement material properties, finishing technique, and seating force play important roles in the ultimate finish line fit.[101,109] Research suggests that finish line form and cement do not significantly affect the fit of cemented crowns.

However, marginal fit varies continuously around the tooth, making an evaluation subjective and arduous.[110–112]

Knife-edged configuration

A knife-edged finish line requires the least amount of tooth reduction. It is sometimes used with a bell-shaped tooth because creation of a heavier margin would require excessive removal of tooth structure. Generally, a knife-edged finish line is not desirable because it is more difficult than other finish lines to discern on a die, and it tends to result in overcontoured restorations. These thin margins are more difficult to wax and cast accurately and are susceptible to distortion under occlusal forces. However, the configuration is commonly used on the mesial aspect of a mesially tipped molar, on the lingual aspects of mandibular molars, and on root surfaces of periodontally involved teeth. It should be noted that excellent results may be obtained with knife-edged finish lines in expert hands coupled with outstanding laboratory support when the gold finish lines are accessible for finishing and are in enamel.[113]

Chamfer configuration

A chamfer is often the preferred finish line for cast gold extra-coronal restorations. It is more conservative than a shoulder finish line and generates less stress at the cement interface.[114] It creates a well-defined and easily identified margin that provides room for adequate thickness of gold without overcon-touring the restoration. Care must be taken not to tilt the bur toward the tooth, because this will increase taper, or away from the tooth, because this will create undercuts.

Shoulder configuration

The shoulder finish line is used primarily when a bulk of material is needed to strengthen the restoration at the margins, such as for all-ceramic or metal-ceramic restorations. It is the least conservative of the finish line types for cast gold. It also has the most critical fit, because there is no bevel present on the restoration that can be burnished against the preparation to reduce marginal opening.

Chamfer or shoulder with a bevel

This design is preferred by clinicians who believe that a beveled margin is easier to detect in an impression and that it makes the margins of the casting more burnishable.

A bevel is recommended for proximal boxes. However, for full cast crowns, the bevel can increase the risk of encroachment on the epithelial attachment.

Contours

The establishment of proper contours of the restoration depends on proper tooth preparation. An overcontoured casting is often the result of an underprepared tooth. If removal of tooth structure is insufficient, the crown must be overcontoured to obtain sufficient thickness of metal. Overcontoured crowns encourage plaque retention, resulting in gingival inflammation[115,116] (Fig 20-10). Even with overcontoured crowns, however, gingival health can be maintained if the patient practices excellent oral hygiene measures.[117]



Fig 20-10 Overcontoured restorations lead to problems. Note the gingival plaque retention and resultant marginal tissue inflammation as a result of facial overcontouring of this all-gold crown.

Wheeler[118] proposed that convexities be created in the gingival third of artificial crowns to deflect food away from the free gingiva. Herlands et al,[119] however, showed that the maximum bulge in the natural crown at its greatest diameter is no more than 0.5 mm greater than at the CEJ and that the impaction mechanism of gingival injury does not occur. The biologic acceptability of undercontouring is often observed when a provisional crown is lost from a prepared tooth for an extended time without adverse effects on the surrounding gingiva.[117] A natural contour (ie, the exact replacement of the axial and proximal morphology) is the desired form for cast gold restorations when the gingival crest is at a normal level.

When the free gingival margin is apical to the CEJ because of recession or surgery, a flattened contour best reproduces the contour of the root surface.[117] Flat contours are recommended occlusal to furcations to allow access for cleaning.

Stein[120] described the contour of a restoration adjacent to the gingiva as the *emergence profile*. He stated that the proximal emergence profiles of all natural

teeth are either flat or concave. This natural contour provides an open gingival embrasure that promotes good oral hygiene.

Occlusion

No restoration, no matter how well crafted, will be successful if it does not function correctly. Satisfactory occlusion is required if the restoration is to achieve adequate function and patient comfort.

Establishing biologically acceptable occlusion starts with careful planning. The teeth opposing the one to be restored (whether in a natural or restored state) should be properly aligned and in the desired occlusal plane. Occlusal surfaces should be well formed. If these conditions do not exist, the opposing dentition should be recontoured or restored, if possible. Failure to do so may seriously compromise the occlusal relationships and, in turn, the future health and function of the involved teeth,[121] as well as the patient's comfort.[122,123]

Existing occlusal relationships should be identified with marking tape in maximum intercuspation position (centric occlusion), centric relation, and excursive movements. Tooth fremitus, radiographic evidence of ligament widening, and wear patterns should be noted. It is essential to determine the precise etiology (eg, physiologic forces with compromised periodontal support or pathologic forces on a normal periodontium) of occlusal traumatism. A broken cusp may result from an overload in working or nonworking movements and is often associated with a large existing restoration. Malpositioned teeth, open contacts, and uneven marginal ridges may warrant additional therapy.[121,123]

Acceptable occlusion has several characteristics. Multiple contact points exist between opposing teeth that come into contact simultaneously during closure. The maximum closure position is referred to as *maximum intercuspation*. Ideally, the facial cusps of the mandibular posterior teeth and the palatal cusps of the maxillary posterior teeth contact the opposing teeth in a fossa or on a marginal ridge so that the occlusal contacts stabilize the teeth in both arches. In some cases, contact of mandibular and maxillary anterior teeth separates the posterior teeth during any eccentric movement of the jaw. This occlusal relationship is referred to as *anterior guidance* or *mutually protected occlusion*.[124,125] Another occlusal relationship, referred to as *group function*, sometimes exists or is created. In this relationship, several teeth on the functional side share equally in the contact during lateral movements of the mandible.[126]

A major benefit of indirect fabrication of a restoration is the ability to form the

wax pattern to the desired occlusal relationships. The wax pattern (and ultimately the restoration) must contact its antagonist(s) in the prescribed contact areas at precisely the instant the other teeth contact. A premature contact or an interference in excursive movements may be uncomfortable, produce loosening or accelerated wear of the restoration or its antagonist, and/or damage the health of the tooth and its supporting structures.[127,128]

Occlusion should be developed with mounted casts on an articulator. For single-unit castings, a simple hinge articulator may be adequate, although it allows accurate reproduction of maximum intercuspation only. Lateral or protrusive interferences must be adjusted in the mouth. In more complicated situations, as for multiple units, use of a more sophisticated semiadjustable articulator may be indicated (Fig 20-11). When casts are mounted on a semiadjustable articulator with a face-bow, the lateral and protrusive movements of the mandible may be simulated with reasonable accuracy.[129]



Fig 20-11 The semiadjustable articulator simulates mandibular movements more closely than the simple hinge articulator.

To verify the accuracy of the relationship of the mounted casts, the patient's occlusal contacts should be checked intraorally with shimstock. If a small discrepancy exists between the patient's occlusal contacts and those on the mounted casts, the casts can be corrected with shimstock, thin articulating paper, and a carving instrument. If the discrepancy is great, it will be necessary to remount the casts. If the casts are mounted accurately and the casting is correctly fabricated, minimal adjustment will be necessary at the placement appointment.

Types of Cast Restorations

The variety of cast restorations ranges from inlays (small intracoronal restorations)

to complete-coverage castings (restorations that cover the entire coronal surface of the tooth) (Fig 20-12). Onlays and partial-coverage castings are hybrids that possess both intracoronal and extracoronal features. The design chosen should be the one that removes the least amount of sound tooth structure while restoring the missing tooth structure and enabling the tooth to withstand functional and parafunctional forces.



Fig 20-12*(a to d)* A variety of designs possible with gold inlay, onlay, partial-coverage, and full-coverage restorations.

Inlay

The gold inlay is a treatment option for moderate-sized Class 1 or Class 2 caries lesions. Although used less frequently than in the past, the inlay has a long history of success. It is not uncommon to see a patient who has multiple inlays that are 30 years old and still clinically serviceable.[9]

Inlays are entirely intracoronal restorations, most commonly with occlusal and proximal extensions (Fig 20-13). When performed with attention to detail and a commitment to skillful execution, inlays can be excellent options for the restoration of caries lesions or defective restorations (Figs 20-14 to 20-16).

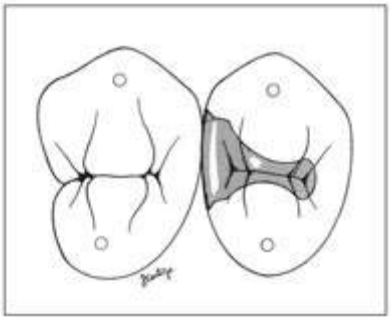


Fig 20-13 An inlay is an intracoronal restoration.



Fig 20-14 Conservative inlays in the maxilla.



Fig 20-15 Occlusolingual inlay.



Fig 20-16*(a)* Conservative disto-occlusal and occlusal preparations for cast gold inlays; *(b)* completed restorations after cementation and finishing.

Preparations should be as conservative as possible to maintain tooth strength. If the occlusal width of the preparation exceeds one-third to one-half the buccolingual intercuspal distance, a restoration offering more protection for the cusps, such as an onlay, should be planned.[40] The occlusal contacts should be entirely on gold or enamel, not on a margin of the restoration.

Block-out technique

After the tooth to be treated has been properly diagnosed and treatment planned as requiring a casting, the operative site is isolated, typically with a rubber dam. The existing restoration and caries are removed using carbide burs in a high-speed handpiece for initial depth and outline form, followed by round burs in a low-speed handpiece or spoon excavators for dentinal caries removal. Obtaining a stain-free dentinoenamel junction is a key objective in this step of cavity preparation. The open

cavity should then be meticulously inspected for cracks, especially for those under cusps or which traverse the tooth mesiodistally or faciolingually. The presence of cracks may require a redesign of the intended restoration to include the coverage of cusps or even the conversion of the preparation and subsequent restoration to a more substantively retentive and resistant restoration, such as an MOD onlay, a three-quarters or seven-eighths crown, or even a full-veneer crown. This flexibility in the operator to change the definitive restorative design demonstrates an appropriate critical-thinking approach to surgical intervention. The next step in the block-out technique is placement of a liner in the deep dentinal areas, followed by filling the cavity nearly to the cavosurface margin with either a composite resin (which may be adhesively bonded) or a reinforced glass-ionomer restorative material. Once the block-out is complete, the operator has a clean start upon which to initiate the following preparation steps. The completed preparation will leave the block-out material only in areas that were previously occupied by dentin, and only for the purpose of avoiding preparation undercuts. All enamel should be supported by dentin, not the block-out material. Figure 20-17 illustrates use of the block-out technique.

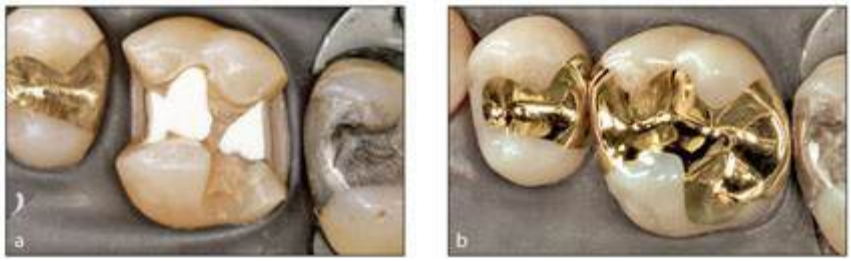


Fig 20-17*(a and b)* A completed preparation for a mesio-occlusodistal lingual (MODL) partial onlay showing the use of a block-out material, which serves as a dentin replacement while allowing enamel to be supported by dentin and facilitates a conservative definitive restoration.

The block-out technique provides several additional advantages when used with indirect restorations: *(1)* allows for a more conservative preparation by blocking out dentin undercuts[130]; *(2)* provides the operator with the opportunity to achieve smooth walls of even depth; *(3)* enhances the visibility of all preparation details; and *(4)* facilitates smoother internal features in the wax patterns and castings, thereby improving the seating and fit of the definitive restoration.[131]

Occlusal preparation

Initial entry is made in the central fossa with a tapered fissure bur to establish the pulpal floor (Fig 20-18a). Theoretically, the ideal occlusal depth is 1.0 mm in dentin or 2.5 mm at the triangular ridges. Depth is ultimately determined by the extent of

existing caries lesions or restorations or the need for additional retention. The occlusal outline is extended mesiodistally along the central groove and stopped just short of the marginal ridge. The bur is held vertically in the long axis of the tooth throughout the preparation. This position and the bur's taper provide the 3- to 5-degree divergence of the facial and lingual walls (total divergence of 6 to 10 degrees) toward the occlusal aspect of the tooth (Fig 20-18b). For two-surface inlays (mesio-occlusal or disto-occlusal), the placement of an occlusal lock or "dovetail" near the marginal ridge will provide a geometry that will resist proximal displacement. The adjacent proximal contact will not provide long-term retention and resistance because teeth move as a result of periodontal ligament compression under normal occlusal loads. When a definite dovetail is not possible because of tooth-structure limitations, an internal retentive feature such as a pin or a slot may be incorporated into the preparation design. The placement of either of these auxiliary features will provide the necessary resistance and retention forms to resist proximal displacement.

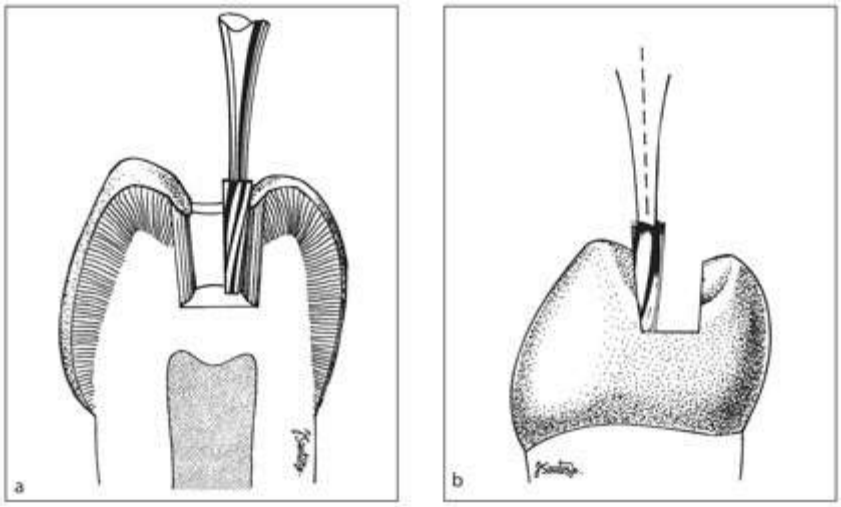


Fig 20-18*(a)* The correct pulpal depth for an inlay is established with a tapered fissure bur. It can be used to create flat floors and well-defined internal angles. *(b)* The tapered sides of the bur are used to help establish the desired occlusal divergence of the walls.

Integral pins and slots

The placement of a pin is a simple two-step process. After identification of an area adjacent to the intact marginal ridge on the pulpal floor, the first step requires the placement of a countersink in dentin with a no. 2 or 4 round bur, to a depth of one-half the bur diameter. This countersink will aid in the placement of the next feature, which is a pinhole. The countersink will also give additional strength to the pin in the finished casting by adding a bulk of gold adjacent to the relatively small integral pin-retentive feature. Pinholes have been historically placed with twist drills of

various diameters, ranging from 0.5 to 0.7 mm, to a depth of 1.5 to 2.0 mm. The parallel pin technique, developed by Shooshan,[132] required the use of nylon impression pins and slightly smaller nylon castable pins, which were then incorporated into the wax pattern and hence the final casting. Because of the difficulty of obtaining nylon parallel pins today, another technique employs the use of a no. 169L bur, which has the advantage of creating a divergent pinhole, which may be captured in the impression by injecting the hole with impression material and ensuring that no air bubbles are trapped. Air may be vented out of the pinhole during impression taking with the use of a small (30-gauge) anesthetic needle, which is placed in the hole while the impression material is simultaneously injected into the hole. The needle is withdrawn after the material has filled the retentive pinhole.

Slots are essentially channels that run parallel to the marginal ridge and dentinoenamel junction and are placed in dentin with a no. 169L bur. Slots will typically measure 1.0 to 2.0 mm in width (faciolingually) and are as narrow as the no. 169L bur at a depth of 1.5 to 2.0 mm. They are similar to pins, but they are bulkier and do not require the countersinks described above for pins (Fig 20-19a). Impression taking for slots may utilize the venting technique described above to capture the details of the retentive feature in the impression[133] (Figs 20-19b and 20-19c).



Fig 20-19*(a)* A mesio-occlusal inlay preparation with a slot-retentive feature placed instead of widening the dovetail. *(b)* Use of a 30-gauge needle to vent out the air while simultaneously injecting impression material into the slot. *(c)* Completed impression of a mesio-occlusal premolar inlay with a slot and a distolingual canine inlay with a Shooshan pin.[132]

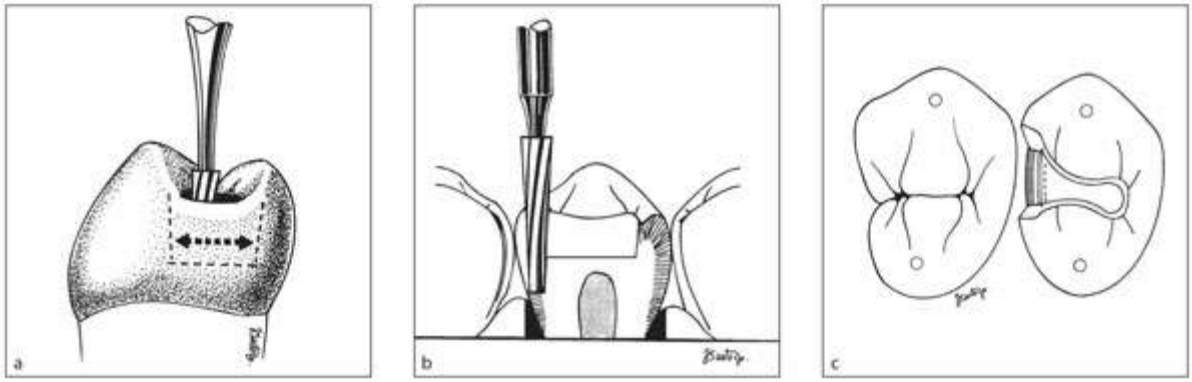


Fig 20-20*(a)* A thin layer of enamel is left on the proximal surface to protect the adjacent tooth while the proximal box is being prepared. *(b)* A flat gingival floor with an axial wall with slight convergence to the occlusal is created. The gingival floor is established occlusal to the gingival tissue, unless otherwise dictated. *(c)* The gingivoaxial line angle is made an even depth into the tooth from the facial to the lingual wall, and the axial wall converges occlusally.

A general guideline for the placement of either slots or pinholes is to place them as far from the primary retentive aspect of the preparation (the proximal box) as possible.[134] It is critical to place integral retentive elements along the line of draw of the preparation, thus avoiding undercuts.

Proximal boxes

The tapered fissure bur is used to create mesial and/or distal proximal boxes. A thin layer of proximal enamel is left to protect the adjacent tooth while the proximal box is formed (Fig 20-20a). The faciolingual dimension is determined by any existing restoration or caries lesion and the relationship of the proximal surface to the adjacent tooth. The gingival floor of the box should have an axial depth of approximately 1.0 to 1.5 mm. Ideally, the gingival extension should be established just apical to the proximal contact (Fig 20-20b). However, the presence of a caries lesion or an existing restoration, or the need for a longer wall to ensure adequate retention, may require extension to a subgingival location. The contour of the axial wall of the box should follow the faciolingual contour of the external surface of the tooth (Fig 20-20c). The box should extend to the facial and lingual borders of the contact area, and the bevels should extend the preparation slightly beyond the box. This extension allows access to the gold margins facially and lin-gually for finishing with a disk.

Refinement

All of the preparation floors and walls should be smooth, all walls except the axial

walls should be divergent occlusally, the axial walls should be convergent occlusally, and internal angles should be well defined (Fig 20-21a; see also Fig 20-18). It is critical that no undercut area exists that could interfere with placement and withdrawal. All cavosurface margins must be distinctly defined (Fig 20-21).

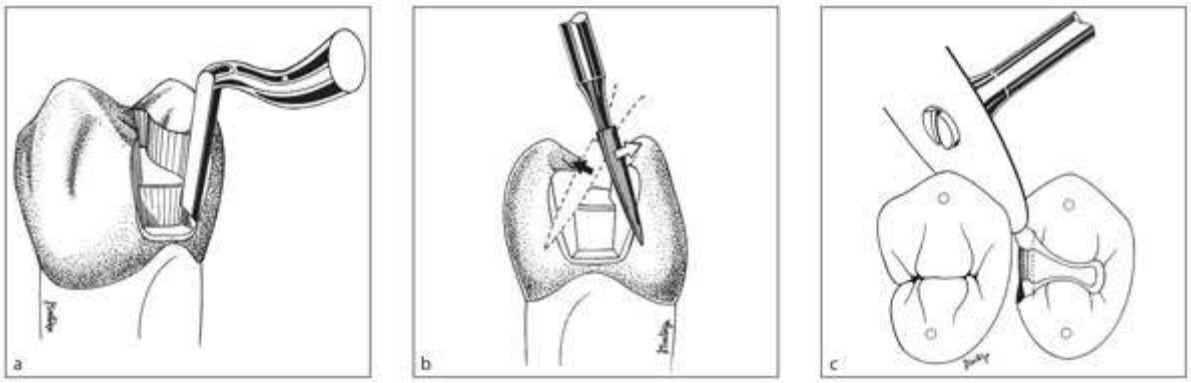


Fig 20-21*(a)* A hand instrument, such as an enamel hatchet, is helpful in smoothing the walls of the preparation. *(b)* The facial and lingual proximal bevels or flares should be established slightly beyond the contact area and blended with the gingival bevel. *(c)* A rotary disk, placed at an angle of 45 degrees, can provide a smooth, flat bevel without undercuts.

Cavosurface bevels

The proximal bevel or flare is established on the facial and lingual walls of the box with a garnet disk, a no. 7901 bur, or a thin, flame-shaped diamond bur (see Figs 20-21b and 20-21c). A finishing bur or diamond bur must be used carefully to avoid developing an undercut in the facial and lingual walls at the faciogingival or linguogingival line angle. An undercut is less likely to occur when a disk is used (see Fig 20-21c). The walls of the preparation should diverge from the gingival floor in the occlusal direction. The proximal bevels should blend smoothly with the gingival and occlusal bevels.

A no. 7901 finishing bur or a thin, flame-shaped diamond bur is used to place 0.5-mm-wide occlusal, proximal, and gingival bevels along the entire cavosurface finish line (Figs 20-22a and 20-22b; see also Fig 20-21b). It may not be possible to place an occlusal bevel if there is a steep incline to the inner cuspal incline. The steep incline may essentially mimic the angle at which the occlusal bevel would normally be placed (Figs 20-22c and 20-22d). A gingival margin trimmer may also be used to place the gingival bevels if access is too limited to use a bur. The bevels should be at an angle of approximately 45 degrees to the external surface of the tooth.

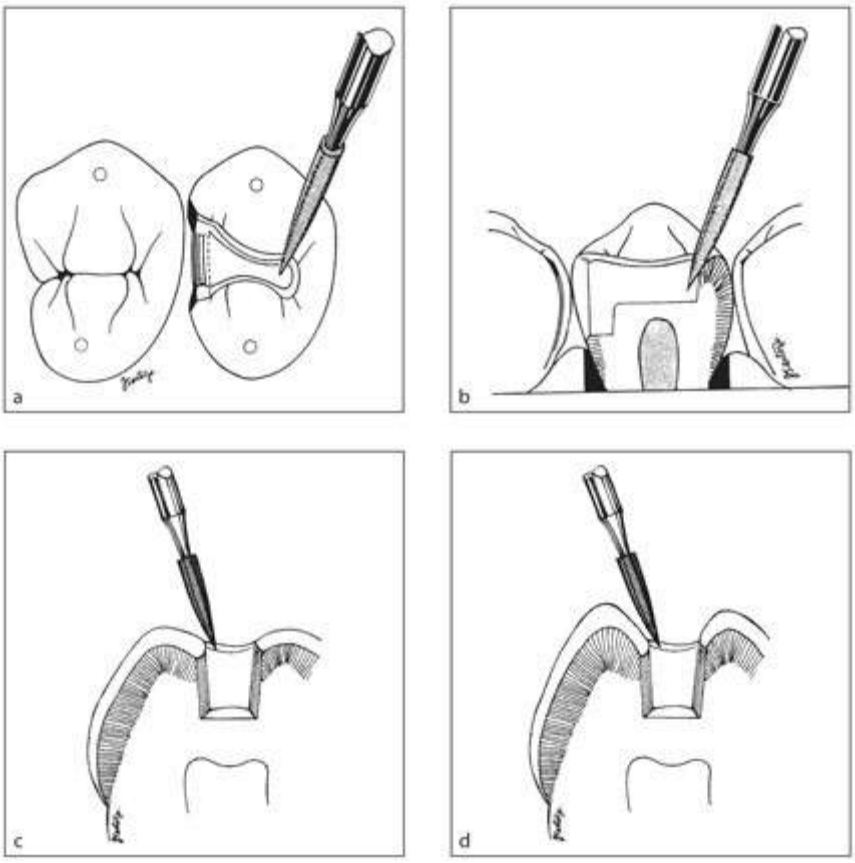


Fig 20-22*(a)* A thin, flame-shaped diamond or finishing bur is used to create a short but distinct bevel at the occlusal finish lines. *(b)* The bevel is extended across the entire occlusal margin and blended with the other bevels. *(c)* An occlusal bevel can be readily placed with a normal or shallow inner cuspal incline. *(d)* It is not possible to place an occlusal bevel with a steep inner cuspal incline because the bevel is essentially parallel to the slope of the cusp.

Retention grooves

If retention grooves are needed to provide additional retention and resistance form, a no. 169 tapered fissure bur is used to place them to bisect the facioaxial and linguoaxial line angles of the box (Fig 20-23). The grooves must diverge toward the occlusal aspect in a facial and lingual direction, and the axial walls should converge occlusally.

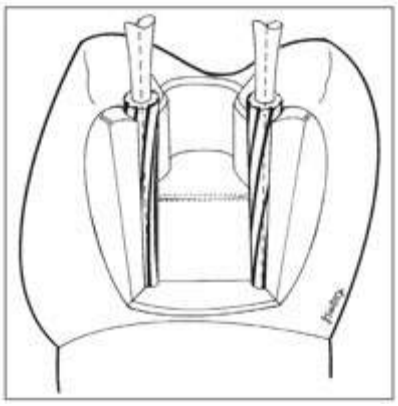


Fig 20-23 The tapered fissure bur (no. 169) is positioned at an angle to give an occlusal divergence for the retentive grooves. The grooves should be at a depth of one-half the diameter of the bur.

Onlay

The onlay is essentially an inlay that covers one or more cusps. A complete onlay covers the entire occlusal surface; a partial onlay covers only a portion of the occlusal surface. The onlay incorporates the principles of both extracoronal and intra-coronal restorations. Although it is generally more conservative than a partial- or complete-coverage crown, it provides the same protection of the remaining tooth structure (Fig 20-24).



Fig 20-24 Onlays on the maxillary premolars provide protection to the facial and lingual cusps while demonstrating minimal display of gold. (Courtesy of Mark Cruz, Monarch Beach, California.)

There are several important features of the preparation. All finish lines are beveled. A bevel or flare creates a second plane designed to allow, with burnishing, close adaptation of the gold to the tooth. A beveled shoulder is used for the centric holding cusps (palatal cusps of maxillary posterior teeth and buccal cusps of mandibular posterior teeth), and a long bevel or chamfer is used for the noncentric cusps. The gingival margin and the facial and lingual walls of the proximal boxes are designed like those for the inlay, with their well-defined bevel or flare. These finish lines are blended to form an uninterrupted finish line around the entire preparation. The gingival floors of the proximal boxes are essentially beveled

shoulders.

The width and depth of the occlusal portion of the preparation and of the proximal boxes are often dictated by the presence of an old restoration and/or a caries lesion. If additional resistance and retention form are needed, retention grooves may be placed at the axiofacial and axiolingual line angles.

A tapered fissure bur is recommended for preparing the outline form because its taper helps to establish the desired total occlusal divergence of 6 to 10 degrees for the internal facial and lingual walls. The axial walls of the proximal boxes converge toward the occlusal surface.

Occlusal preparation

The initial entry is made in the central fossa to a depth of approximately 1.0 mm into dentin (total depth of approximately 2.5 mm in the tooth). In some cases, it may be necessary to extend some portions of the preparation to a greater depth because of carious dentin or a previous restoration or for additional retention. The occlusal outline form is extended by moving the bur laterally, cutting with the side of the bur. The occlusal outline form should be as conservative as the caries lesion or old restoration permits. The bur is kept in the long axis of the intended path of insertion so that the taper of the bur provides the desired 3- to 5-degree divergence for each internal cavity wall.

Proximal boxes

The boxes are created on the proximal surfaces. The facial and lingual walls should exhibit a combined divergence occlus-ally of 6 to 10 degrees from each other as was provided in the occlusal area of the preparation (Fig 20-25a). The faciolingual dimension is likely to be determined by the presence of a restoration, caries lesion, and/or the relationship of the proximal surface to the adjacent tooth (Fig 20-25b). The bevels at the facial and lingual cavosurface angles will extend the preparation slightly beyond the proximal contact area so that the margins of the restoration will be accessible for finishing with a disk (Fig 20-25c).

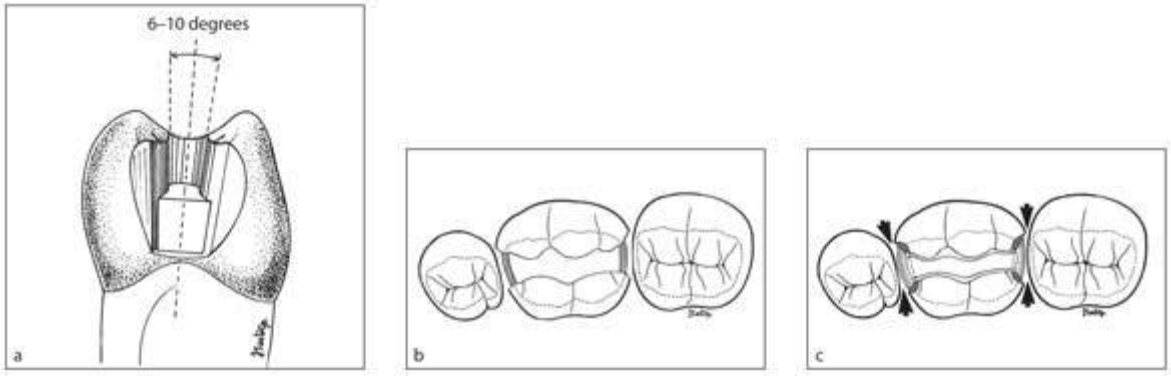


Fig 20-25*(a)* Ideally, opposing walls diverge 6 to 10 degrees for the inlay and onlay preparation. *(b)* The proximal box is extended to or slightly beyond the contact area. Bevels will provide the desired proximal clearance. *(c)* After the bevels are placed, there is access for completing the preparation's finish lines and the margins of the restoration.

Cuspal reduction

A carbide or diamond bur is used to reduce the cusps. Depth cuts of 1.5 to 2.0 mm are made for the centric (vertical holding) cusp(s), and cuts of 1.0 to 1.5 mm are made for the noncentric cusp(s) (Fig 20-26a). A bur with a measured diameter is used to gauge the depth of the cuts. The side of the bur is held parallel to the cuspal inclines to make the depth cuts. After the depth cuts are placed, a uniform reduction of the cusps that parallels the general anatomical contours of the occlusal surface is made. The cuspal heights are reduced to the full extent of the depth cuts (Figs 20-26b and 20-26c). Reduction for the centric holding cusps is generally greater than that for the noncentric cusps because less thickness of the restoration is needed to withstand occlusal forces exerted against a noncentric cusp.

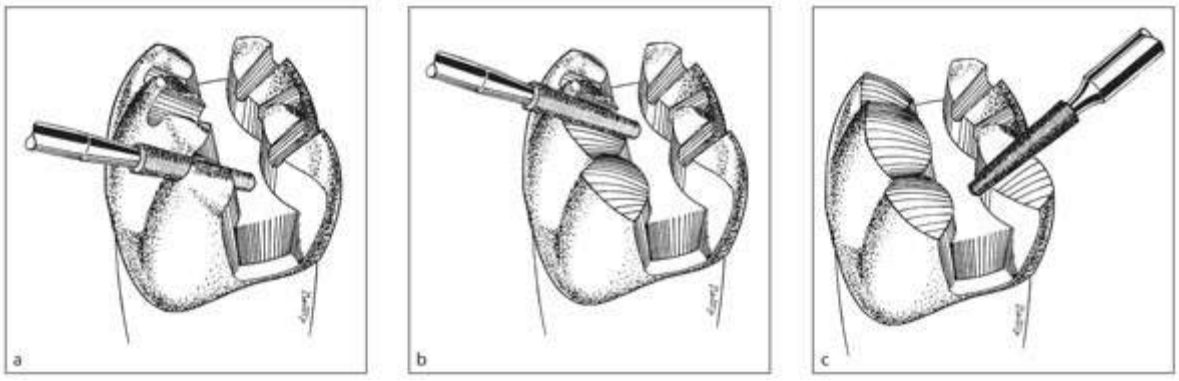


Fig 20-26*(a)* A bur of known diameter is used to establish depth cuts to guide the correct reduction of the cusps: 1.5- to 2.0-mm-deep cuts on centric holding cusps and 1.0- to 1.5-mm-deep cuts in noncentric cusps. *(b)* The buccal cusps are reduced in accordance with the occlusal anatomy of the tooth. *(c)* The lingual cusps of the mandibular teeth require less reduction because they are not holding cusps.

Shoulder preparation

A shoulder is prepared on the external surface of the centric cusp to provide a band of metal (ferrule) to protect the tooth. The bur is held parallel to the external surface of the tooth to prepare a shoulder about 1.0 mm in height and 1.0 mm in axial depth (Fig 20-27). The finish line should extend gingivally at least 1.0 mm beyond any occlusal contacts. The occlusoaxial line angles are rounded. There must be adequate (1.0- to 1.5-mm) clearance in all eccentric mandibular movements.

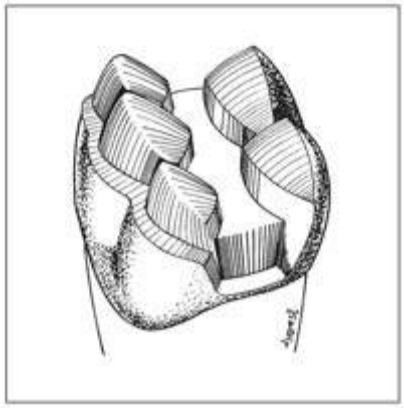


Fig 20-27 The shoulder should have precise line angles.

Noncentric cusp

A chamfer or long bevel may be used instead of a shoulder on the noncentric cusp(s). The bur is positioned at an angle of approximately 45 degrees to the axial surface (Fig 20-28). This provides a ferrule effect for additional protection of the cusp.

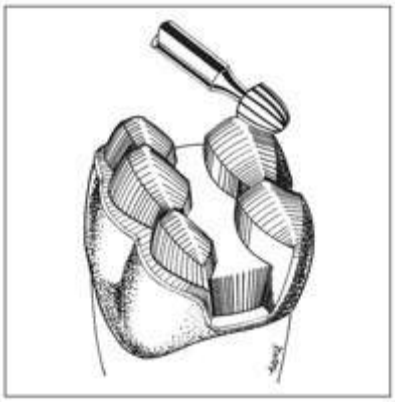


Fig 20-28 A barrel-shaped bur can be used to create the chamfer on the noncentric cusp(s).

Gingival bevel

A smooth and distinct bevel is established on the gingival margins with a no. 7901 finishing bur; a thin, flame-shaped diamond bur; or a gingival margin trimmer. If a

gingival margin trimmer is utilized, it should be positioned so as to create a 45-degree angle to the blade. This bevel should be approximately 0.5 to 1.0 mm in width and at an angle of approximately 45 degrees to the external surface of the tooth.

Shoulder bevel

A 0.5- to 1.0-mm bevel is placed on the shoulder with a no. 7901 or fine diamond bur. This bevel is blended with the proximal bevels. Any corners or sharp angles at the junction of the various bevels and across the occlusoaxial line angles are eliminated (Fig 20-29).

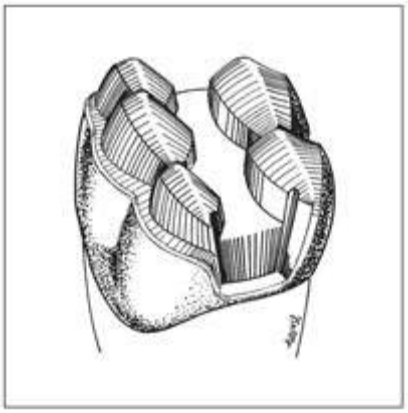


Fig 20-29 Finished onlay preparation. Internal angles are precise, occlusal line angles are rounded, the walls have the correct taper, the grooves are correctly positioned, and the finish lines are smooth and continuous.

Proximal bevels

The proximal bevel or flare is established with a garnet disk; a fine, flame-shaped diamond bur; or a no. 7901 bur. Creation of an undercut during beveling at the faciogingival or linguogingival line angles must be avoided. Divergence is established from the gingival floor occlusally. The proximal bevels should blend smoothly with the gingival bevel and the buccal and lingual bevels.

Retention grooves

If retention grooves are needed, they can be placed in both proximal boxes. A no. 169 bur is used to bisect the facioaxial and linguoaxial line angles. The grooves must diverge toward the occlusal aspect faciolingually and converge toward the occlusal aspect axially to be aligned with the internal path of insertion (Fig 20-30).

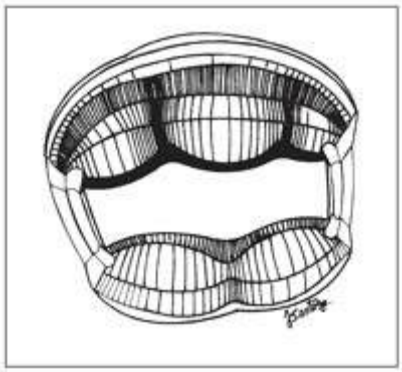


Fig 20-30 The retention grooves are placed at the linguoaxial and facioaxial line angles but do not undermine the enamel.

Partial-coverage crown

The partial-coverage crown, or partial veneer crown, covers only a portion of the outer circumference or axial surfaces of the tooth and completely covers the occlusal surface. For instance, a three-quarters crown has three of the four axial surfaces covered by the restoration, usually leaving the facial axial surface unprepared, except at its occlusal extent; this is to preserve the facial surface for esthetics. A seven-eighths crown, usually used for maxillary molars, covers three and one-half of the four axial surfaces, usually leaving the mesiofacial half of the facial surface of the tooth unprepared. Leaving part of the external surface of the tooth uncovered offers several potential benefits. It conserves tooth structure and avoids potential insult to the periodontium adjacent to the unrestored tooth surface.[95] The uncovered tooth surface allows for pulp testing.[128] Preservation of the facial surface eliminates the need to match the shade of the adjacent teeth with a ceramic material. Because it is not necessary to veneer the casting with a tooth-colored material, the laboratory procedures are simplified.[135]

Retention and resistance of the partial veneer crown are provided by a combination of extracoronal and intracoronal features. The extracoronal retention is created by opposing axial walls on the mesial and distal surfaces that have a combined convergence toward the occlusal of approximately 6 to 10 degrees. This is supplemented with grooves or boxes in the proximal walls that provide not only added retention but also resistance to lingual displacement. A slight overlay of the facial cusp protects it from fracture and provides some resistance form.

The extensions of the proximal and facial portions of the preparation can vary in design according to the specific needs. If the tooth is located so that the facial cusps are readily visible, esthetic concerns may dictate a modification of the standard

design to conceal the gold. In such cases, the mesial proximal wall is extended toward the facial surface only far enough to barely break contact. Reduction of the mesial incline of the mesiofacial cusp of molars and the mesial incline of the facial cusp of premolars is limited. The distal incline is reduced more to provide protection and the ferrule effect. This more esthetic design is shown in Fig 20-31.

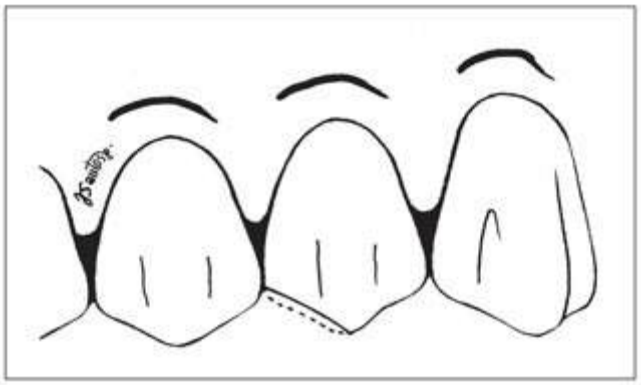


Fig 20-31 Limited reduction of the mesiofacial incline of the facial cusp and greater reduction of the distofacial incline of a maxillary tooth provide some protection to the cusp while limiting the display of gold in a partial-coverage restoration.

Because partial-coverage crowns are infrequently placed on anterior teeth, this section focuses on the posterior teeth.

Occlusal reduction

Depth cuts are made by laying the side of the bur (of measured diameter) against the cuspal inclines (Fig 20-32a) and reducing them to the desired depth (Fig 20-32b). The total reduction of the centric cusp should be 1.5 to 2.0 mm, and that of the non-centric cusp should be 1.0 to 1.5 mm. The remaining occlusal surface is reduced, but the general anatomical contours are maintained (Fig 20-32c). As previously mentioned, less reduction may be desirable in some areas for esthetic reasons. The placement of an occlusal channel and the facial bevel are discussed later.

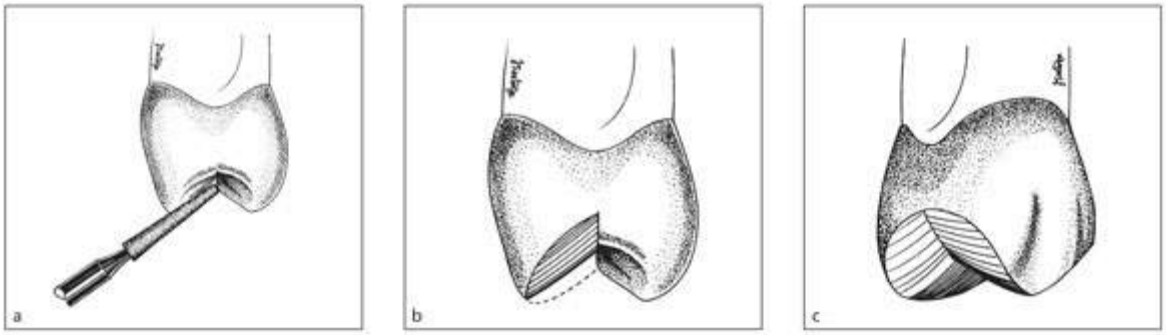


Fig 20-32*(a)* The diamond bur is held at the same angle as the natural slope of the cusp to create an even occlusal reduction for partial-coverage crowns. *(b)* Removal of 1.0 to 1.5 mm of tooth

structure from the noncentric cusp serves as a guide to reducing the rest of the occlusal surface. *(c)* The occlusal reduction has followed the original contours of the occlusal surface.

Lingual reduction

The axial wall of the lingual surface is reduced with a round-ended tapered diamond bur. Close attention must be paid to the desired path of insertion to establish parallelism. A two-plane reduction of the tooth is needed to maintain natural contours. Because of the anatomical differences in the lingual contours of the maxillary and the mandibular teeth, the second plane on the lingual surface of a maxillary tooth will be more pronounced. The gingival portion of the lingual surface should have a 3- to 5-degree convergence occlusal to the path of insertion (Fig 20-33), and the second plane should be offset about 30 degrees.

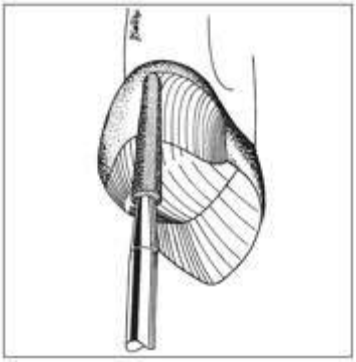


Fig 20-33 The round-ended tapered diamond bur is very effective in creating the proper taper and in creating a chamfer finish line.

Proximal reduction

The proximal surfaces are reduced in one plane. A 3- to 5-degree taper is established from the finish line to the occlusoaxial line angle. Space limitations may require use of a thin tapered diamond bur initially until enough space has been created to use a round-ended tapered diamond bur. The round-ended diamond bur has a more appropriate shape to create a chamfer finish line (Fig 20-34).



Fig 20-34 The lingual contours and finish line should be carried into the proximal surfaces. Lack of access may require use of a thinner diamond bur initially.

Finish lines

The junction of each proximal wall and the lingual wall is blended so that there is a smooth transition. This procedure is especially important at the finish line. The gingival finish lines are placed slightly coronal to the gingival tissue if possible. The presence of caries lesions or existing restorations may alter the level of finish line placement, but, regardless of where the finish line is located, the transition should be smooth and well defined. For the buccal finish line of partial-coverage crowns for maxillary teeth, several different configurations may be used, balancing the need for resistance form and facial esthetics (Fig 20-35).

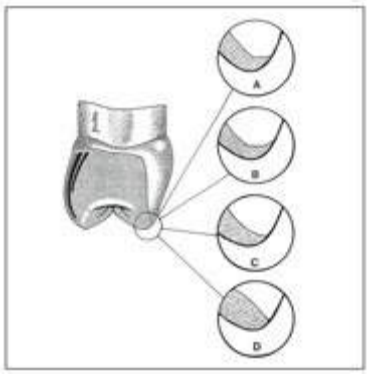


Fig 20-35 Several occlusobuccal finish lines for maxillary partial-coverage crowns—flat bevel (A), contrabevel (B), and knife-edged (C)—are acceptable finish lines. A bevel is not absolutely necessary with adequate enamel support; however, the finish line in D would compromise finish and resistance form with unsupported enamel, making a bevel necessary. (Redrawn from Shillingburg et al,[136] after Ingraham et al[137] and Richter and Ueno.[84])

Proximal grooves

The grooves are initiated in the proximal areas with a no. 170 bur (or a no. 169 bur for a small tooth). The grooves are located as far facially as possible without undermining the facial enamel. It may be helpful to mark the proposed location of the grooves before they are prepared (Fig 20-36). Because the grooves must have a path of draw compatible with each other and with the axial walls, their angulation must be carefully planned before they are begun. As a general rule for posterior teeth, the grooves should be parallel to the long axis of the tooth (Fig 20-37a). The axial walls of the two grooves should converge occlusally. Failure to align the grooves correctly will result in a preparation that does not have an acceptable path of insertion (Figs 20-37b and 20-37c).

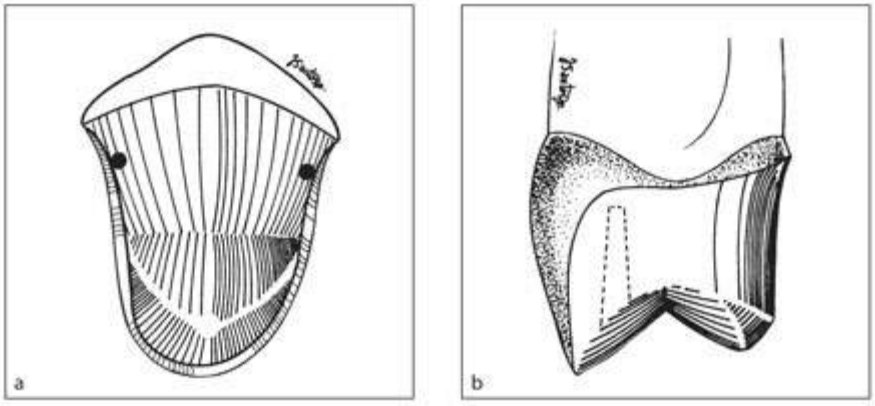


Fig 20-36*(a)* Before the proximal grooves are prepared, their desired location is marked with a pencil or an indentation made by a bur. This aids in obtaining the correct faciolingual location. *(b)* The angulation of the proximal groove is equally important, so visualization is critical.

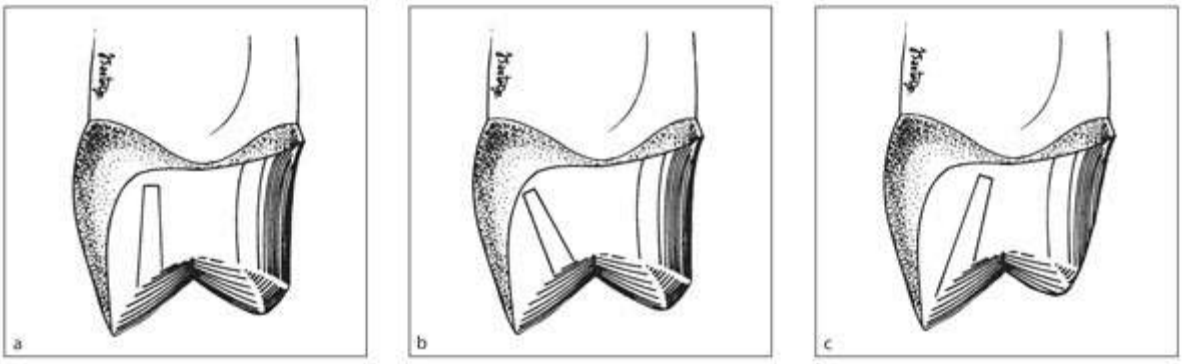


Fig 20-37*(a)* Alignment of the proximal groove with the long axis of the tooth provides the best compatibility with the rest of the preparation design. *(b)* This proximal groove is angulated too much toward the lingual surface and will not provide adequate resistance to lingual rotation. *(c)* This proximal groove is angulated too much toward the facial surface, creating an undercut in relation to the lingual surface of the preparation, thus interfering with the path of insertion.

The axial depth of each groove is made equal to or slightly greater than the diameter of the no. 170 bur. After the first proximal groove is cut, the groove on the opposite side of the preparation is cut in the same fashion so that it aligns with the first groove and with the axial walls (Fig 20-38). The grooves may be enlarged, and the internal walls may be left rounded or more acutely refined to form boxes. In many cases, a box form is present after the removal of an existing restoration. Existing boxes may be modified and incorporated in the preparation in lieu of grooves (Fig 20-39).

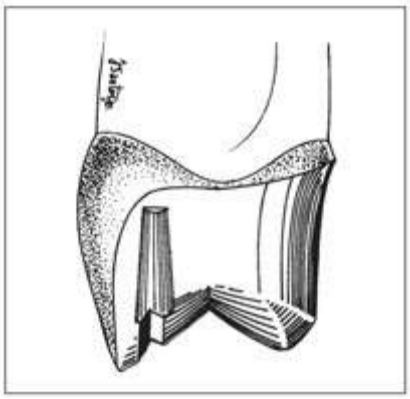


Fig 20-38 Properly designed and placed proximal groove.



Fig 20-39 A groove is located on one proximal surface and a box on the other. Both meet the requirements for occlusal divergence of the facial and lingual walls and convergence of the axial wall toward the opposite proximal wall. The groove has a pronounced lingual wall to enhance resistance to lingual displacement.

Occlusal channel

In those circumstances where esthetics is a significant consideration, typically for maxillary premolars, less reduction of the facial cusp is desired to reduce the display of gold. This often necessitates that the preparation of the lingual slope of the facial cusp is less than 1.0 mm, with a minimal facial cusp bevel. For this situation, the resistance form and strength of the restoration can be enhanced by preparation of an occlusal channel. A channel may already exist if an occlusal restoration has been removed. If not, it may be prepared. A flat-ended tapered carbide or diamond bur is used to cut the channel or to remove undercuts in an existing channel. The channel allows the space for a “staple” of thicker metal in the restoration that resists lingual displacement and helps the restoration resist deformation under pressure (Figs 20-40 and 20-41).

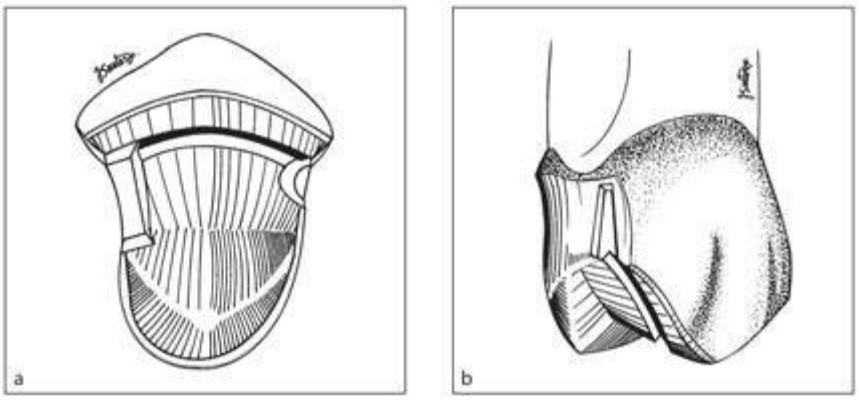


Fig 20-40*(a)* The occlusal channel generally parallels the contour of the facial surface rather than cutting straight across the tooth to the opposite wall. *(b)* Note the shape of the occlusal channel as it comes across the occlusal surface to connect to the proximal groove. Also note the small facial bevel.



Fig 20-41 In the mandible, the centric holding cusp requires a facial design similar to that of the onlay preparation.

Facial bevel

Placement of the facial finish line differs in the maxillary teeth and the mandibular teeth. Because the maxillary facial cusp is usually the noncentric cusp, only a 1.0-mm layer of metal, or less, and a short bevel are required to protect the cusp (see Fig 20-40). Placement of a shoulder and a bevel are recommended for the facial cusp in mandibular teeth to help restorations withstand the forces on centric cusps. A shoulder extending 1.0 mm occlusogingivally and 1.0 mm deep axially is placed into the facial surface across the facial cusp with a straight fissure bur held parallel to the external surface of the tooth (see Fig 20-41). A 0.5- to 1.0-mm-wide bevel is placed with a fine diamond or finishing bur. The finish line should be placed gingival to any occlusal contacts.

Final refinement

All the sharp external angles of the preparation are rounded. Sharp angles make it more difficult to pour the stone into the impression without bubbles. Even if the die is poured without bubbles, it has fragile edges that are easily abraded during laboratory procedures. After all the angles of the preparation are rounded, the surfaces are smoothed with a fine-grit diamond bur. Figure 20-42 illustrates some completed preparations.

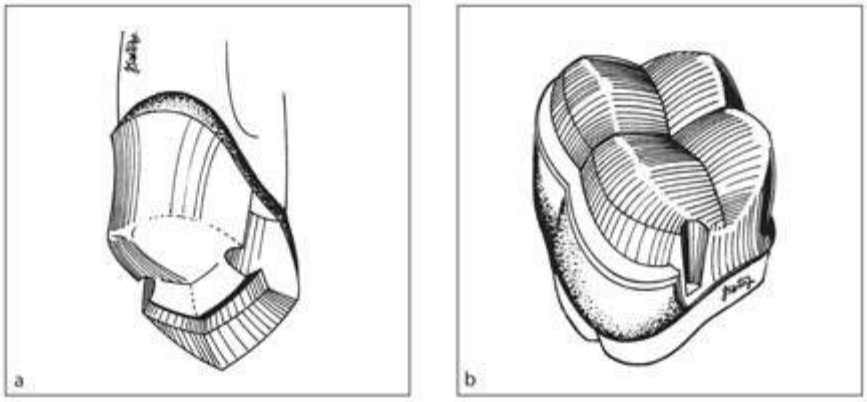


Fig 20-42 Preparations for partial-coverage crowns require the same refinement of walls, floors, and bevels as are needed for inlay and onlay preparations. *(a)* Maxillary premolar preparation. *(b)* Mandibular molar preparation.

Complete-coverage gold castings

As its name implies, the complete-coverage, or full veneer, casting includes coverage of the entire coronal portion of the tooth. Extensive loss of tooth structure is the most common indication for a complete-coverage restoration. For esthetic reasons, this restoration tends to be limited to molars. Because the restoration involves the entire circumference of the tooth, control of occlusal and proximal relationships allows improvements in occlusion and proximal contacts. Correction of tooth positioning is sometimes possible.

Retention is provided primarily or entirely by the extracoro-nal walls. A complete-coverage crown is the most retentive of the casting designs.[138,139] One or more grooves or boxes can be added to the preparation if additional retention and resistance form are needed.

Proximal reduction

A thin tapered carbide or diamond bur is placed at either the facial or lingual embrasure and used to cut proximal tooth structure toward the opposite embrasure (Fig 20-43). The bur should be extended cervically to the desired location of the gingival finish line. Unless a caries lesion, an old restoration, or the need for

additional retention/resistance features dictates otherwise, the gingival finish line should be established at least 0.5 mm supragingivally. The reduction must be at the expense of the tooth being prepared to avoid damage to the enamel of the adjacent tooth. The reduction, when completed, should taper occlusally 3 to 5 degrees toward the opposing proximal wall. The opposing wall is prepared in the same manner. Once enough space has been created to permit access, a round-ended tapered diamond bur is used to complete the proximal reduction and place a gingival chamfer.

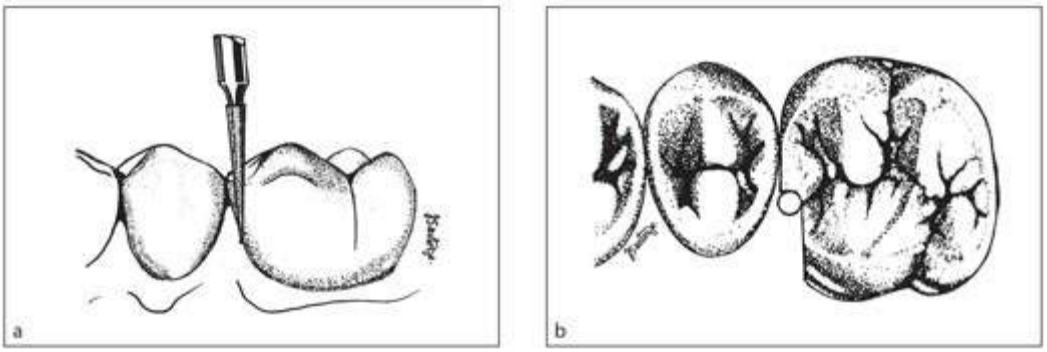


Fig 20-43*(a)* Use of a thin, flame-shaped diamond bur is recommended to initiate the proximal reduction of the preparation for a complete-coverage restoration. *(b)* The reduction is kept within the original contours of the tooth so that abrasion of the adjacent tooth is avoided.

Occlusal reduction

A series of depth cuts is made on the occlusal surface with a diamond or fissure bur. Both the facial and lingual cuspal inclines are cut to a depth of at least 1.5 mm to allow an adequate thickness of metal for strength and wear. An even reduction that follows the original anatomical contours allows the development of the appropriate occlusal anatomy in the restoration.

Facial and lingual reduction

The facial and lingual surfaces are each reduced in two planes to allow reproduction of normal contours in the restoration. The first plane, extending occlusally from the gingival finish line, should have a 3- to 5-degree taper to the path of insertion (Fig 20-44a). The second plane should be angled at 30 to 45 degrees to the first plane to allow desired facial and lingual contours to be reestablished (Figs 20-44b to 20-44d). This second plane should begin at the occlusal third of the wall for the centric cusp and the occlusal fourth for the noncentric cusp. A round-ended tapered diamond bur is used to make these reductions. A modified conservative full-coverage preparation, involving supragingival margins on both lingual and buccal aspects, has

shown excellent long-term results.[113]

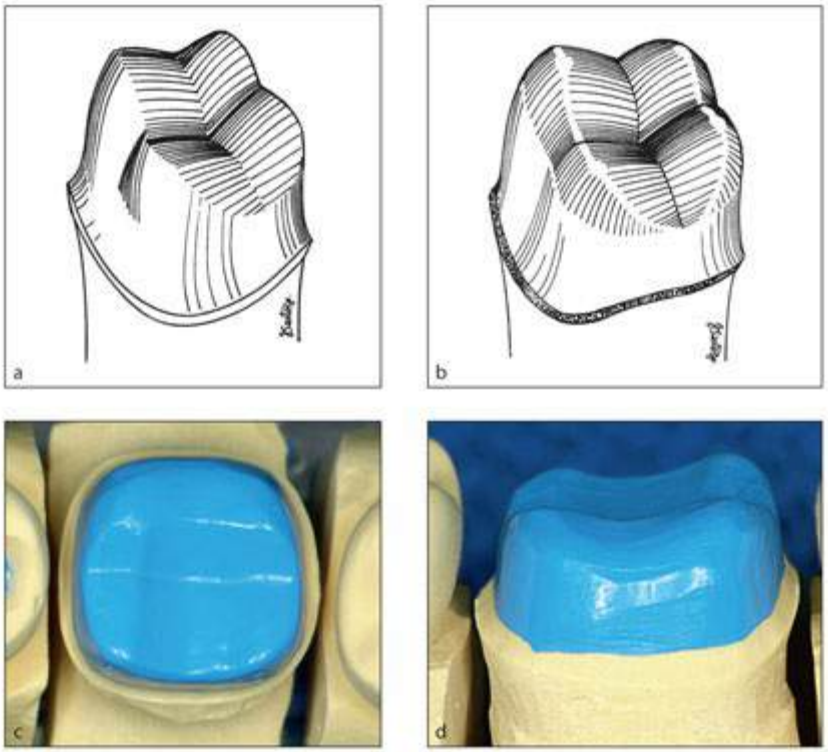


Fig 20-44*(a)* Reduction of the axial walls for a complete-coverage restoration is identical to that for partial-coverage crowns, except the facial wall is also included. At this stage, the facial and lingual walls are each in one plane. *(b)* The second plane is added to the facial and lingual walls. The second plane should be at a 30- to 45-degree angle to the long axis and involves the occlusal third of the centric holding cusp and the occlusal fourth of the noncentric cusp. *(c)* Occlusal view of the die of a full crown preparation (with blue spacer applied) for a mandibular right first molar. *(d)* Facial view of the preparation showing the second plane of reduction.

Refinement

The parts of the preparation are blended to eliminate indefinite areas along the finish line, irregularities in the walls, and sharp corners. A finish line that is distinct around the whole preparation must be established. The facioproximal and linguoproximal line angles often need to be rounded and distinctly defined. All of the sharp corners of the preparation, such as the occlusoaxial line angles, are rounded. The surfaces of the preparation should be relatively smooth. The preparation is viewed carefully from more than one angle to ensure that there are no undercuts within any wall, between the walls, or in relationship to an adjacent tooth.

Try-In

Before the casting is fitted to the prepared tooth, it should be adjusted to fit the

master die. Research has demonstrated that use of several layers of die spacer can increase the retention of cemented crowns and improve the fit of cemented restorations.[140–142] The intaglio surface of the casting is inspected under magnification for bubbles or other imperfections,[143,144] which can be removed with a small round bur. The die is checked for defects or abraded areas. The casting should not be forced onto the die. If the casting does not seat easily, the die should be sprayed with a disclosing medium, and the crown should then be gently reseated. The disclosing medium should mark areas that are binding on the inside of the casting. These areas can be relieved with a bur (no. 2 or no. 4 round bur). This process is repeated until the casting is fully seated.

Once the casting is fully seated on the die, the interproximal contacts are checked and necessary adjustments made. This is best accomplished on a solid cast, especially if there are multiple castings. The occlusion in maximum intercuspation is adjusted until the restoration holds shimstock equally as well as the adjacent natural teeth. The location and size of the occlusal contacts are marked with articulating paper. A finishing bur, stone, or abrasive disk is used to make any modifications needed. After maximum intercuspation contacts on the restoration are similar to maximum intercuspation contacts on other teeth in the quadrant or arch, if the type of cast articulation is fairly accurate, contacts in excursive movements are marked and adjusted to remove excursive interferences. The internal surface of the casting may be carefully air abraded with aluminum oxide, avoiding the margins, in preparation for the clinical try-in. Air abrasion provides a dull, matte finish. Any area that binds during intraoral seating of the casting creates a bright, burnished mark. If these steps are accomplished with precision before the patient arrives, minimal chair time should be required for adjustments.

Once adjustments have been made on the cast, the casting is tried on the prepared tooth to determine the fit. Adjustments to the proximal contacts are made if needed. The internal surface is inspected for shiny spots. The shiny spots are adjusted, and the casting is reseated. The casting is removed and rein-spected for new shiny spots. This process may need to be repeated several times until the casting is fully seated.

An alternative method to seat castings employs disclosing media. Disclosing media include chloroform and rouge, disclosing pastes or waxes,[145] and impression materials.[146] Whatever the choice, the medium is placed on the internal surface of the casting, and it is then seated. The casting is removed and inspected for abrasion of the disclosing medium that allows the gold to show through. These areas are adjusted as needed, more disclosing medium is placed, and the casting is reseated. This process is continued until the casting is fully seated.

The fitting process is completed when the margins are flush against the finish lines of the preparation and there is no binding when the restoration is seated. The casting should fit passively in place. A tight fit usually means the casting is not fully seated. Use of a die spacer medium, painted onto the die to within 0.5 to 1.0 mm of the finish line prior to wax-pattern fabrication, will usually simplify the process of fitting the casting.[146]

Marginal Finishing

Two options exist to complete the finishing of accessible margins of cast gold restorations: *(1)* precementation finishing and *(2)* postcementation finishing.

Precementation finishing

In this option, marginal finishing is done prior to cementation to adapt and polish margins that have already been determined to be satisfactory. The goal is to develop a margin that is adapted to the preparation finish line; extends to, but not beyond, the finish line; and blends with the contours of the tooth.[147] The casting must be fully seated before an attempt can be made to adapt and finish the margins.

Margin location determines whether margin finishing can be done on the tooth. Subgingival and/or interproximal margins are very difficult to reach without damaging the soft tissue, bone, or the tooth itself. For these margins, all finishing must be done on the die. Easily accessible margins can be finished while the casting is seated on the tooth (Fig 20-45). If the margin is tightly adapted but slightly overcontoured or undercontoured, a white stone or abrasive disk can be used in a low-speed handpiece to reduce the protruding surface, whether it is gold or enamel. When finished, the gold margin should be flush with the tooth structure, and adjacent contours of the restoration should be continuous with natural tooth contours.

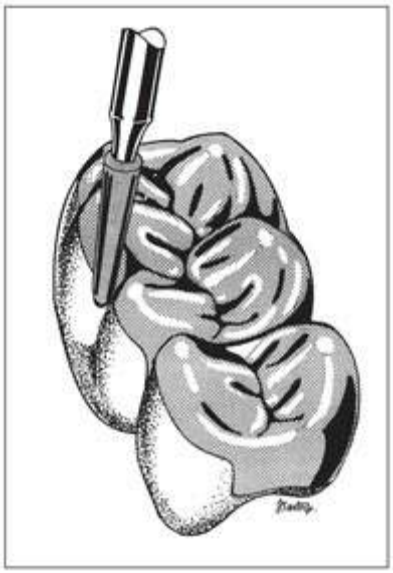


Fig 20-45 A fine diamond bur or finishing stone is held perpendicular to the margin and moved parallel to the margin to reduce small discrepancies in the marginal area.

The best instruments for finishing accessible surfaces are flexible paper disks.[148] Sandpaper disks should not be used to try to close an open margin if the casting is not fully seated. This usually leads to a visible cement line. Rather, the gold casting should be completely seated so that it is brought level to the cavosurface margin. If there is a slightly overextended margin after the casting is completely seated, the use of disks allows the operator to remove the slight vertical excess of gold without creating any gaps or cement lines.[102] The stone or disk should be rotated from the metal to the tooth or parallel to the margin, but never from tooth to gold. Running the rotary instrument from gold to tooth ensures a better seal and prevents the gold margin from being peeled away from the tooth interface. Rubber points or fine-grit abrasive disks may be used to produce a high luster if access permits. Care must be exercised to avoid damage to the soft tissue or abrasion of the tooth surface.

Marginal adaptation can sometimes be improved by hand burnishing, both on the die and on the tooth prior to cementation. Well-controlled pressure is applied with a beavertail or ball burnisher, held adjacent to and moved parallel to the gold margin (Fig 20-46). The burnisher should not be placed directly on the margin. Pressure should be applied with a back-and-forth motion (parallel to the margin) that moves slowly closer to the margin. Because a small area of the burnisher is contacting the gold, a great deal of pressure (force per unit area) is applied. The casting is stabilized during the burnishing process to ensure that it does not change position. It is also critical to have good finger rests to ensure that the burnisher does not slip off the tooth and injure the soft tissue or the tooth itself if the burnishing is done in the

mouth. Some additional benefit may be achieved by burnishing the margins of the casting on the tooth during the cementation procedure, before the cement sets.[149]

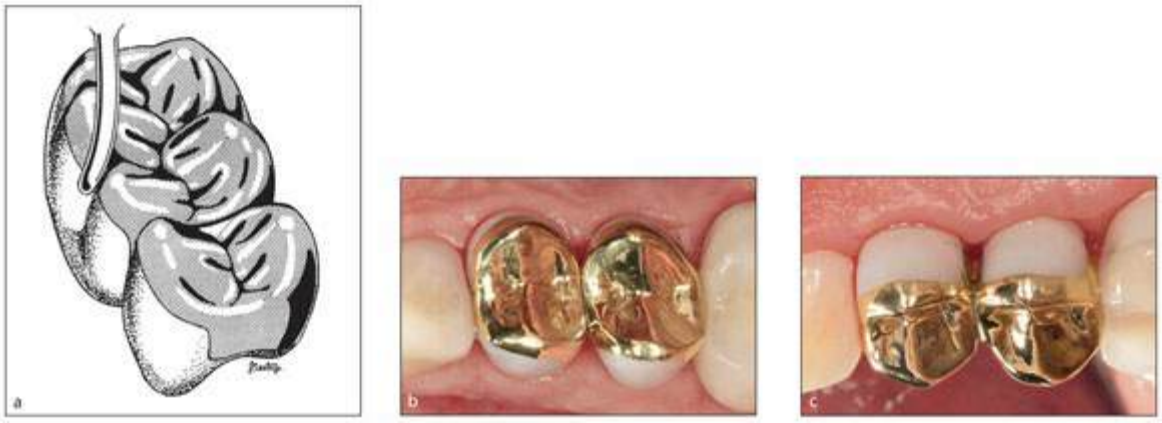


Fig 20-46 *(a)* Pressure is applied with the side or tip of the burnisher, which is moved parallel to the marginal area to adapt it more tightly to the tooth. *(b and c)* Mandibular partial-coverage restorations. Note the well-adapted margins that have been burnished.

Cementation

The final step in the process is luting or cementation of the casting. Cement failure has been shown to be the second[150] or third[151] most important factor in the failure of cast restorations. A thin layer of the luting agent (cement) is placed in the casting and on the walls of the preparation, and into any auxiliary retentive features, and the casting is then seated on or in the tooth. Complete seating of the casting should be verified. The cement hardens within a few minutes, and the excess is removed. The luting agent fills the gap between the casting and the tooth to provide retention and to minimize leakage at the margin. Some luting agents, such as zinc phosphate cement, provide retention entirely from frictional resistance to displacement. Others, such as the adhesive resin cements, bond to the tooth and/or the casting to provide adhesion in addition to frictional resistance. It appears that resin cements or resin-modified glass-ionomer cements provide superior retention and have better physical properties than more traditional cements.

However, it is essential to understand that no cement will compensate for inadequate preparation resistance and retention form in the long term. Therefore, when sound preparation design principles can be used, the choice of cement is of little clinical relevance.[152,153] But clinical situations often vary from the ideal, so cement can be of real importance in these cases of compromised resistance/retention form, as suggested by a recent case report of two occlusal onlays, placed on a minimally retentive preparation and cemented with resin-modified glass-ionomer

cement, that were successfully retained for 6 years.[154] One has to remain prudent as not to draw hasty conclusions that cement will override improper preparation. Sound principles should prevail unless proven otherwise by long-term clinical research.

Postcementation finishing

In this option, a casting that has been determined to be well fitting is cemented and subsequently finished and polished on the tooth itself. The same steps used in the precementation technique are employed after the cement is fully expressed from the casting. This technique affords the operator the advantage of developing a high polish and ideal marginal adaptation simultaneously, without the need to repolish after cementation.

An understanding of the advantages and limitations of each type of cement is relevant and is presented in the following section.

Selection of a luting agent

Options for luting agents today are zinc phosphate, glass-ionomer, polycarboxylate, resin, and resin-modified glass-ionomer cements.[155] Resin cements and resin-modified glass-ionomer cements have gained popularity in recent years, with much of the current research in the area of cements focused on these two materials.

According to surveys,[156,157] there has been a significant shift in routine cement use from zinc phosphate, zinc polycarboxylate, and conventional glass ionomer to resin and resin-modified glass-ionomer cements. More than two-thirds of North American practices have elected resin cements and resin-modified glass-ionomer cements as their first choice. Purportedly, they are superior to zinc phosphate in terms of retention, strength, and solubility and are comparable in terms of film thickness.[158–160]

The type of cement used does not seem to influence the final marginal adaptation. The ease of manipulation and strict adherence to the protocol for each specific luting agent have a more significant impact on the ultimate strength and solubility properties, which, in turn, affect the clinical performance of the definitive restoration.[152,161] Because the indications and procedures for each luting agent are different, they are discussed separately.

Zinc phosphate cement

Zinc phosphate cement consists of a powder containing 90% zinc oxide and 10% magnesium oxide that is incorporated into a liquid containing approximately 67% phosphoric acid.[31] Among the many luting agents available, zinc phosphate has the longest record (more than 100 years) of successful use.[162] It has good compressive strength and low film thickness, and it is relatively easy to manipulate.[31] Excess cement is easily removed after it has set. Relatively high compressive and tensile strengths make it a good choice for long-span fixed partial dentures.[163] In a study of zinc phosphate cement samples taken from castings that had been in service for up to 48 years, the cement was found to have maintained a stable chemical structure.[164]

Zinc phosphate cement is slightly soluble in oral fluids, allows relatively high levels of microleakage,[165] and is sometimes associated with postoperative sensitivity.[166,167] Despite these drawbacks, it is not unusual to see patients with gold castings that were cemented more than 30 years ago with zinc phosphate cement. In a study of eight restorations that had been in service for 22 years despite positive microleakage tests, no evidence of carious tooth structure, sensitivity, or pulpal degeneration was observed.[168]

Zinc phosphate cement should be mixed on a glass slab that has been chilled, preferably to just above the dew point to avoid water condensation that could contaminate the mix and weaken the set cement. Powder should be mixed into the liquid in small increments over a large area of the glass slab. This method dissipates the heat released during the exothermic setting reaction and provides a longer working time. The slower the mix and cooler the glass slab, the longer the working time. A cool glass slab also allows more powder to be incorporated into the liquid, resulting in improved mechanical properties.[169]From a practical standpoint, refrigerated glass slabs (4şC) always induce water condensation, but the coldness of the slab still allows the incorporation of more powder to the mix, which compensates for the water contamination and leads to improved compressive strength.[170]

Powder is added until the mixed material adheres to the spatula to form a 1-inch string when the spatula is lifted from the glass slab. The mixing should be completed within 90 seconds after initiation. If the mixing is prolonged beyond approximately 90 seconds, the hardening of the cement caused by the setting reaction may be confused with having achieved the proper powder-liquid ratio.[171] Its relatively long working time makes zinc phosphate cement a good choice when multiple castings are luted at the same time. However, because of a lack of a bond to tooth structure and the mixing-technique sensitivity, zinc phosphate is used less frequently now.

Glass-ionomer cement

Traditional glass-ionomer luting cement is a modification of the glass-ionomer restorative material. It consists of a powder containing aluminum fluorosilicate glass particles that are incorporated into a liquid containing polyalkenoic acids. In some products, these acids are freeze-dried and included in the powder that is mixed with water.[172] It bonds to tooth structure by formation of ionic bonds to hydroxyapatite crystals in enamel and dentin.[172] It is speculated that glass ionomers have the potential to inhibit caries because of fluoride release, but the critical level of fluoride release over time to prevent caries is well beyond the release rate of current glass-ionomer luting cements.[173,174] Glass-ionomer luting cement has a low film thickness (20 to 25 μm) and better mechanical properties than zinc phosphate cement.[31,175,176] However, a glass ionomer may take a significant amount of time before reaching ultimate strength.[177,178] Glass-ionomer cements may be hand-mixed but are also available in preweighed capsules that allow mixing in an amalgamator. Empirical reports of excessive postoperative sensitivity associated with glass-ionomer luting agents have not been borne out in clinical studies.[179–182] Glass-ionomer cements appear to be more technique sensitive than zinc phosphate cements because the surface of the preparation needs to remain moist before cementation and the cement at the margin of the restoration must be isolated from moisture but not desiccated after cementation.[152]

Glass-ionomer cement is very sensitive to early moisture contamination. If exposed to external moisture during the setting reaction (usually 5 minutes), the setting reaction is interrupted, resulting in a cement with high solubility and poor mechanical properties.[183] Therefore, glass-ionomer cement should not be used unless contamination can be prevented. As with glass-ionomer restorative materials, dehydration is also a problem.[184] Excess cement should not be removed from the margins for at least 5 minutes after the crown is seated. Leaving a bead of excess cement in place protects the underlying layers of cement from moisture contamination.[185] Because glass-ionomer cement bonds to the tooth surface, excess cement is somewhat difficult to remove once set, particularly in the interproximal areas.

When glass-ionomer cement is mixed by hand, the pre-measured powder is incorporated into the premeasured liquid in bulk as quickly as possible. A chilled slab has been shown to significantly increase the working time for glass-ionomer cements, as it has for zinc phosphate cements.[186] Most manufacturers now offer an encapsulated form of glass-ionomer luting cement; this results in properly

proportioned cement that is thoroughly mixed in a few seconds. Glass-ionomer cement has a relatively short working time, so a single mix should be limited to luting no more than two or three units. If additional units have to be luted, a new mix of cement should be used.

Resin-modified glass-ionomer cements

Resin-modified glass-ionomer cements were introduced in the late 1980s. Like resin cements and traditional glass-ionomer cements, resin-modified glass-ionomer cements are modified restorative materials. The combination of resin and glass-ionomer chemistries overcomes some of the problems with moisture contamination and dehydration experienced with glass-ionomer cements[187] and eliminates many of the steps necessary for resin cements.[155] In addition, they release fluoride, providing potential anticaries activity.[188] Resin-modified glass-ionomer cements have physical and mechanical properties between those of conventional glass-ionomer and resin cements.[176] Their adhesive properties to tooth structure are similar to those of traditional glass ionomers,[189] and they have adequately low film thicknesses.[161,190] A recent study showed that both resin cements and resin-modified glass-ionomer cements met the 25-μm film thickness requirement when tested 2 minutes after mixing.[191] However, it is recommended to finish the seating procedure within 10 seconds of the manufacturer's working time to ensure complete seating. A recent study demonstrated that the powder-liquid formulation, as opposed to the paste-paste equivalent, exhibited a significantly better bond to dentin than any of the self-adhesive resin cements.[160]

Resin-modified glass-ionomer cements offer little to no postcementation sensitivity.[192–194]

Finally, resin-modified glass-ionomer cements absorb fluids during and after setting, causing a net expansion.[195,196] One anecdotal clinical report and two in vitro studies[197–199] have linked early resin-modified glass-ionomer formulations with all-ceramic crown fractures. The current generation of resin-modified glass-ionomer cements has apparently overcome this deficiency.[200,201]

Resin luting cements

Resin cements patented since 2002 come either with a separate self-etching bonding agent, such as Panavia F (Kuraray) and Multilink Automix (Ivoclar Vivadent), or with an incorporated self-etching primer, such as RelyX Unicem (3M ESPE) and

Maxcem (Kerr). These cements have shown low microleakage, high shear bond strength, and adhesive properties to tooth structure. However, they have other additional challenges. First, they are significantly more technique sensitive than traditional luting agents. Most require multiple time-consuming steps and strict moisture control. In addition, resin cements are very difficult to remove from crown margins once they are completely set. Removal of cement flash is made difficult by the relative inaccessibility of crown margins, irregular root topography, and the clear or tooth-colored nature of many resin cements. For these reasons, many clinicians do not use resin cements routinely for posterior applications. However, some circumstances dictate the use of resin cement. Because of their adhesive properties, a resin cement is the material of choice when luting a crown with minimal retention and resistance features. In addition, resin cements must be used when bonding all-ceramic etchable, noncore crowns. As is the case with most dental materials, a clear understanding of the advantages and disadvantages, as well as clinical indications, is essential when making a material selection for a given circumstance.

Resin cements are especially designed for use with bonded ceramic restorations, but they may also be used with cast restorations. They are generally supplied as a dual-paste system in which the two parts are mixed just prior to the cementation process. Polymerization may be initiated by a chemical catalyst or by a combination of a chemical catalyst and a light-activated catalyst, but for cast restorations, chemically cured cements are indicated. Resin cements differ from resin composite restorative materials primarily in their reduced filler content[202]; reduction of the filler content leads to better flow and reduced film thickness, which are desirable properties for a luting cement.[203] Some resin cements contain fillers that are capable of providing fluoride release,[204] although there is questionable therapeutic effect because of the low levels and short duration of fluoride release.[205]

Resin cements have the best mechanical properties of all of the cements.[176,206,207] They are virtually insoluble in oral fluids and have the highest compressive and tensile strengths of all the cements. They also exhibit less microleakage[208] than other luting agents. If used in combination with a dentin adhesive system, they bond to tooth structure and to some metals.[169] For this reason, as previously stated, they are often recommended for less-retentive preparations.[209–211]

There are several potential problems associated with resin cements. There is great variation in the mechanical properties and handling characteristics among resin cements,[190] which can cause confusion for clinicians. The film thickness tends to be

greater than that of other cements,[173,190,212] so incomplete seating of the casting could be a problem. This is especially true when a dentin adhesive is used, because without due care, it can pool in the internal angles of the preparation. Resin cements require the most clinical steps if used with an adhesive system, so there is more chance for an error in technique to occur. There are no long-term clinical studies to determine if the high retention values and low microleakage are long lasting in castings cemented with resin cements.

Polycarboxylate and zinc oxide–eugenol cements

Polycarboxylate and zinc oxide–eugenol cements are no longer widely used with cast restorations. Their properties are generally inferior to those of the cements previously discussed. Historically, some clinicians have used them if a tooth has had a history of sensitivity. They have also been used as temporary luting agents and with stainless steel crowns.

Preparing the tooth for cementation

Although some materials are more adversely affected by the presence of contaminants than others, no material fares well if used in the presence of oils, debris, saliva, blood, or other significant contamination.[213,214] It is important to ensure that the area is cleaned, free of excess moisture, and well isolated. Bleeding and other significant sources of contamination must be well controlled. Temporary cements leave a layer of debris on the dentin surface[215] that should be removed before cementation. This may be accomplished with hand instruments, pumice, detergents, and/or cleaning agents.

Some clinicians recommend the additional step of disinfecting the preparation with chlorhexidine, ethylenediaminetetraacetic acid (EDTA), or benzylkonium. The rationale is that bacteria are a primary cause of postoperative sensitivity,[71,72] so disinfection of the dentin surface will lower the number of microbes and thus reduce one cause of sensitivity. Another approach is to use desensitizing agents, usually including hydroxyethyl methacrylate (HEMA). These agents can be applied immediately before the final impression. They essentially disinfect the dentin and seal the tubules. Their use does not seem to interfere with crown retention, regardless of the type of luting cement used.[216–218]

In addition to cleaning and isolating the preparation, specific additional steps are recommended to prepare the tooth to receive certain cements. Preparation surface

texturing obtained either by rotary instruments or sandblasting with 50-μm aluminum oxide has been shown to increase crown retention significantly, regardless of the choice of cement.[202,219,220]

Zinc phosphate cement

Because zinc phosphate cement exhibits the most microleakage of any cement$_{190}$ and has been associated with postcemen-tation sensitivity, several methods have been recommended to prevent sensitivity by "sealing" the dentin prior to cementation. Copal varnish has a long history of use for this purpose, with anecdotal reports of success. However, copal varnish does not disinfect the dentin, and its effectiveness has limited duration.[221,222]

More recently, use of dentin primers and adhesives to seal the dentin before cementation has resulted in reports of decreased sensitivity. However, a study has noted a marked (42%) decrease in retention with the application of a resin-based sealer prior to cementation.

Glass-ionomer cements

For glass-ionomer cements, the tooth should be clean and slightly moist.[179] No cavity varnish should be placed that might prevent bonding of this luting agent to the tooth surface. The area should be well isolated to prevent moisture contamination during the luting process and for several minutes following the seating of the restoration.

Resin-modified glass-ionomer cements

One of the advantages of resin-modified glass-ionomer cements is that the cementation procedure is fairly simple. Multiple bonding steps are not necessary, and no special preparation of the tooth surface is performed, other than to make it clean and slightly moist. Removal of excess cement is relatively easy.

Resin cements

Most resin cements have corresponding dentin adhesive systems that are applied immediately before cementation. In most cases, the preparation is etched and a primer and adhesive are placed. Each adhesive system has specific instructions for its use that must be followed precisely to obtain the best results. Failure to follow the specific directions for both the adhesive and the luting resin can mean the

difference between success and failure. It must be emphasized that excellent isolation is a necessity when a resin cement is used.

A small amount of the cement should be mixed and observed before cementation procedures are started to confirm that the cement will set. Some self-curing resin cements have short shelf lives and lose their ability to polymerize.

Preparing the casting for cementation

After adjustments have been made and the external surface of the casting has been polished, the internal surface of the casting should be air abraded with aluminum oxide to produce a uniform, slightly roughened finish. Margins must be avoided or protected when the internal surface is air abraded. The casting should then be cleaned to remove any contaminants, such as polishing compounds. Ammonia, detergents, or various other cleaning solutions may be used in an ultrasonic bath, or the casting can be cleaned with a steam cleaner.

If an adhesive resin cement is used, additional retention can be obtained by tin plating the inside of the casting.[223,224] A number of inexpensive tin-plating systems that can be used to deposit a thin layer of tin on the surface of the gold alloy are commercially available. This simple process can be done in the laboratory or operatory in a few minutes.[223] A newer generation of metal-surface treatments, including the Rocatec[225] (3M ESPE), Silicoater$_{226}$ (Heraeus Kulzer), and CoJet[227] (3M ESPE) systems, all produce a coating that has been shown to significantly increase bond strength of resin to metal. Yet another alternative that has shown promising results is the use of an adhesive system that has an alloy primer.[228]

Seating the casting

With zinc phosphate and glass-ionomer cements, the cement is mixed and placed in the casting. Some clinicians also like to place a layer of the cement on the walls of the preparation for intracoronal restorations. The casting should be half-filled with cement, and all of the margins should be covered. Once the casting is seated with finger pressure, the patient is instructed to bite on an orangewood stick or other seating instrument. If a rubber dam is being used for isolation, two orangewood sticks may be tethered together with dental floss to allow the patient to close on the sticks and not the bow of the rubber dam retainer (clamp). The instrument is moved up and down and then side to side. This technique, called *dynamic seating*, results in more complete seating of the casting.[229] The margins are checked with an explorer

to determine that complete seating has been accomplished. For small intracoronal castings, such as occlusal and two-surface inlays, a sharpened stick may be placed with heavy force on the casting, along the line of draw of the preparation, with light mallet tapping. This latter technique requires continued tapping with the mallet until no further cement expresses from the marginal areas. The clinician needs to understand how the cement choice relates to finishing technique.

The optimum finishing technique takes place on the die with a final finish applied while the cement is setting. An extremely high luster may be placed on the cemented castings using ribbed (not webbed) polishing cups and a sequence of no. 4 flour of pumice, 15-µm aluminum oxide, and 1-µm aluminum oxide powders, either wet or dry. Use of these agents, particularly dry, can generate a significant temperature increase, and they must be used with light, intermittent pressure.[148]

Recently, a pertinent study explored the marginal adaptation of cast gold inlays with three cements—zinc phosphate, resin-modified glass-ionomer, and self-adhesive modified resin cement—and three finishing techniques.[149] Finishing the casting before and during cementation led to the least marginal opening for all three cements.

The excess cement must be removed with proper timing and care. If removal is attempted before the cement is set, thin strands of cement may be dragged from under the margins, thus creating voids. The beads of extruded cement have been shown to protect the cement at the margin from moisture contamination.[185] Therefore, it is important to check the cement to ensure that it has set before the excess is removed. Excess cement may be removed with an explorer or curette directed parallel or away from accessible margins, and floss or yarn is used to remove interproximal excess cement. The gingival crevice should be flushed with water frequently to remove any loose particles. As a final step, an assistant should supply a continuous stream of air to each margin while the clinician gently reflects the free gingiva with an explorer to check for any remaining cement. Cement left subgingivally can cause tissue inflammation. A knot is tied in the floss and then the floss is pulled back and forth through the gingival embrasure to remove loose pieces of cement from the interproximal areas.

As discussed previously, there are some distinct advantages of adhesive resin luting cements in certain situations, but their use requires different seating procedures than those used for zinc phosphate and glass-ionomer cements. For each resin-based cement, manufacturer instructions should be followed closely. The internal surface of the casting should be air abraded, tin plated, or both to increase retention. Procedure guides accompanying the resin cement should be adhered to.

Removal of excess adhesive resin cement from exposed areas of the tooth and adjacent teeth can be a bit difficult, so a large amount of excess cement should be avoided. The adhesive and luting materials are mixed and applied according to the manufacturer's instructions, and the casting is seated using procedures similar to those used for other cements. Setting times for these materials vary, and the time to remove excess cement is critical, so, again, instructions must be followed closely.

The occlusion is reevaluated and final adjustments are made, if necessary. An easy technique to adjust high spots is to use a fine egg- or football-shaped carbide bur, followed by abrasive polishers on the adjusted area, to bring the casting back to a smooth surface and high polish. The patient is informed of the possibility of postcementation sensitivity and that a minor additional adjustment of the occlusion may be necessary. The patient is instructed in proper oral hygiene measures to ensure prevention of caries and periodontal problems.

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